

Limiting absorption principle and scattering matrix for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity

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Abstract

We study scattering theory for a class of matrix-valued Schrödinger operators on a one-dimensional lattice. In our model, the free Hamiltonian is given by a variation of the discrete Laplacian, and we consider a matrix potential satisfying a condition of exponential decay. After studying the resolvent and calculating the Green's function, we calculate the fibration of the scattering operator with respect to the energy.

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Introduction

We study scattering theory for a class of matrix-valued Schrödinger operators on $\mathbb{Z}$ of form

$$H = H_0 + V,$$

where the free Hamiltonian H_0 is an energy shift of the discrete Laplace operator composed with the left multiplication by some invertible matrix A and the potential V is a self-adjoint matrix multiplication operator obeying an exponential growth condition (cf. Section 1 for a detailed definition). This model is widely used in the context of low-energy phenomena in solid-state physics.

The novelty of this model is that, if the matrix A that is not a scalar multiply of a unitary matrix, the dimension of the scattering matrix varies depending on the energy. Such models have recently been of interest, for example, in [APRR24], although there, in a similar model, just the case of a potential V supported in a single point has been studied.

A construction of the corresponding scattering matrix has been done in the case $A = \mathbf{1}$ in [BFGS22] and [BFNS24], and for the general case in [BS25]. Even though these papers establish some properties of the constructed scattering matrices, a formal proof that the matrices defined by them do, in fact, coincide with the fibration of the scattering operator with respect to the energy of the free Hamiltonian has not been provided. The present paper aims to close this gap.

Structure of the paper:

In Section 1, we formally introduce the model.

After this, we show that the function $\sigma(H_0) \ni E \mapsto S^E$, constructed in [BS25], coincides with the fibration of the scattering operator $S = \Omega_-^* \Omega_+$ with respect to the energy E . Our proof of this result is loosely based on the ideas presented in [RS79, Chap. XI.6]. However, since, contrary to our discrete model, [RS79, Chap. XI.6] considers a model on a continuous space, we had to make some profound adaptations. The outline of our proof is as follows:

In Section 3, we analyze the resolvents of the operators H_0 and H , which leads to formulas for the Green's function (Theorem 3.1.3) and the limits of the resolvent when approaching the spectrum (Theorem ??).

1 The model

Given $L \in \mathbb{N}$ and a invertible normal matrix $A \in \text{GL}_L(\mathbb{C})$, we consider the free Hamiltonian H_0 on $\mathcal{H} := \ell^2(\mathbb{Z}, \mathbb{C}^L)$ defined by

$$(H_0\phi)(n) := A^*\phi(n+1) + A\phi(n-1).$$

We now consider a map

$$V : \mathbb{Z} \rightarrow \mathcal{M}_{L \times L}(\mathbb{C})$$

with

$$V(n) = V(n)^* \quad \forall n \in \mathbb{Z},$$

satisfying the condition

$$\sum_{n \in \mathbb{Z}} \|V(n)\| \tau_A^{|n|} < \infty, \tag{1}$$

where

$$\tau_A := \eta_A + \sqrt{\eta_A^2 - 1} \quad \text{with} \quad \eta_A := \|A\| \cdot \|A^{-1}\|.$$

Then, for $\phi \in \mathcal{H}$, we set

$$(V\phi)(n) := V(n)\phi(n)$$

and consider the interaction Hamiltonian

$$H := H_0 + V.$$

2 Generalized eigenfunctions

2.1 Basic properties

In this section, we consider the generalized eigenfunctions of H_0 and H respectively, and establish some of their properties.

Definition 2.1.1. Let $N \in \mathbb{N}$. For a matrix-valued function $\phi : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N}(\mathbb{C})$, by abuse of notation, we define

$$(H_0\Phi)(n) := A\Phi(n+1) + A\Phi(n-1)$$

and

$$(V\Phi)(n) := V(n)\Phi(n)$$

and set

$$H := H_0 + V.$$

Further, given $K \in \{H_0, H\}$ and an energy $E \in \mathbb{C}$, we write $\mathcal{G}_N(K, E)$ for the set of generalized eigenfunctions of E to the generalized eigenvalue E , i.e. for the set of functions $\phi : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N}(\mathbb{C})$, such that

$$K\phi = E\phi.$$

We write

$$\mathcal{G}(K, E) := \mathcal{G}_1(K, E)$$

for the space of generalized eigenfunctions with values in $\mathbb{C}^L \cong \mathcal{M}_{L \times 1}(\mathbb{C})$.

Remark 2.1.2. Since, for any $n \in \mathbb{N}$, a solution $\phi \in \mathcal{G}_N(K, E)$ is uniquely determined by $\phi(n)$ and $\phi(n+1)$, one has

$$\dim \mathcal{G}_N(K, E) = 2LN,$$

so especially

$$\dim \mathcal{G}(K, E) = 2L.$$

Further, given a function $\phi \in \mathcal{G}_N(K, E)$, for every $v \in \mathbb{C}^N$, one has $\phi \cdot v \in \dim \mathcal{G}(K, E)$.

The idea for finding generalized eigenfunctions of the free Hamiltonian is to consider functions $\phi : \mathbb{Z} \rightarrow \mathcal{M}_{L \times L}(\mathbb{C})$ of the form

$$\phi(n) := Z^n$$

for a matrix $Z \in \text{GL}_L(\mathbb{C})$. Then, one has

$$H_0\phi = A(Z + Z^{-1})\phi.$$

To find a generalized eigenfunction of H_0 with energy E , one therefore has to find a matrix $Z \in \text{GL}_L(\mathbb{C})$ with

$$A(Z + Z^{-1}) = E\mathbf{1}.$$

Such matrices can be chosen depending holomorphically on the energy E in the following subsets of the complex plane:

Definition 2.1.3. We set

$$\mathcal{S} := \{z \in \mathbb{C} : |\text{Re } z| > 2a_L\}$$

and, for $\sigma \in \{\pm\}$, we define

$$\mathbb{C}_\sigma := \{z \in \mathbb{C} : \sigma \text{Im } z > 0\}$$

and

$$C_\sigma := \mathbb{C}_\sigma \cup \mathcal{S} \cup [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A)).$$

Proposition 2.1.4. ([BS25, Prop. A.3.2]) *For $\sigma \in \{\pm\}$, there exist open subsets $O_\sigma \supseteq C_\sigma$ and holomorphic functions*

$$z_{\sigma,k} : O_\sigma \rightarrow \mathbb{C}^\times, \quad k = 1, \dots, L,$$

such that, for the functions

$$Z_\sigma := \text{diag}(z_{\sigma,1}, \dots, z_{\sigma,L}) : O_\sigma \rightarrow \text{GL}_L(\mathbb{C}),$$

the following hold:

(a) *For every $E \in O_\sigma$, one has*

$$A(Z_\sigma(E) + Z_\sigma(E)^{-1}) = E\mathbf{1}.$$

(b) *For every $E \in C_\sigma$, one has*

$$\|Z_\sigma(E)\| \leq 1$$

with strict inequality holding for $E \in \mathbb{C}_\sigma \cup \mathcal{S}$.

(c) *For every $E \in \overline{\mathbb{C}_\sigma} \setminus (\sigma(2A) \cup \sigma(-2A))$ and $k = 1, \dots, L$, one has*

$$0 \leq -\sigma \text{Im } z_{\sigma,k}(E)$$

with strict inequality holding for $E \in \mathbb{C}_\sigma$.

(d) *The set O_- is the complex conjugate of O_+ and for every $E \in O_+$, one has*

$$Z_+(E) = Z_- (\overline{E})^*.$$

From now on, let O_σ , $z_{\sigma,k}$ and Z_σ , $\sigma \in \{\pm\}$, $k \in \{1, \dots, L\}$, be as in the above proposition. We then can define generalized eigenfunctions of the free Hamiltonian:

Definition 2.1.5. (cf. [BS25, Def. 2.1.4]) For $\sigma, \tau \in \{\pm\}$ and $E \in O_\sigma$, we define the function $\phi_\tau^{E,\sigma} : \mathbb{Z} \rightarrow \mathcal{M}_{L \times L}(\mathbb{C})$ by

$$\phi_\tau^{E,\sigma}(n) := Z_\sigma(E)^{\tau n}.$$

Then, by construction, one has

$$H_0 \phi_\tau^{E,\sigma} = E \phi_\tau^{E,\sigma}$$

(see [BS25, Lem. 2.1.5(a)]).

Given these generalized eigenfunctions of the free Hamiltonian, we want to construct generalized eigenfunctions of the interacting Hamiltonian. Here, we have the following result:

Theorem 2.1.6. ([BS25, Cor. 2.2.3, Thm. 2.2.2(e)]) *Let $\sigma, \tau \in \{\pm\}$. Then the following hold:*

(a) *For every $E \in C_\sigma$, there exists a unique function $u_\tau^{E,\sigma} \in \mathcal{G}_L(H, E)$ such that*

$$\lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} u_\tau^{E,\sigma}(n) = \mathbf{1}.$$

Further, for every $n \in \mathbb{Z}$, the map

$$C_\sigma \ni E \mapsto u_\tau^{E,\sigma}(n)$$

is continuous and holomorphic on C_σ° .

(b) *For every $E \in C_\sigma$, one has*

$$\lim_{n \rightarrow \tau\infty} u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n) = 0.$$

(c) *If $F \subseteq C_\sigma$ is a compact subset, for every $N \in \mathbb{Z}$, one has*

$$\sup_{E \in F, \tau n \geq N} \|u_\tau^{E,\sigma}(n)\| < \infty.$$

The previous discussion allows us to make the following definition:

Definition 2.1.7. Let $K \in \{H_0, H\}$, $\sigma, \tau \in \{\pm\}$ and $E \in C_\sigma$. We set

$$m_{K,\tau}^{E,\sigma}(n) := \begin{cases} \phi_\tau^{E,\sigma}(n) & \text{if } K = H_0 \\ u_\tau^{E,\sigma}(n) & \text{if } K = H, \end{cases} \quad n \in \mathbb{Z}.$$

For our analysis of these generalized eigenfunctions, the following set will play a crucial role:

Definition 2.1.8. (cf. [BS25, Def. 2.1.7]) For $E \in C_\sigma$, $\sigma \in \{\pm\}$, we set

$$J^E := \{k = 1, \dots, L : |z_{\sigma,k}(E)| = 1\}.$$

(We will see in the following proposition that, for $E \in C_+ \cap C_-$, this definition is independent of σ .) Further, let $P^E \in B(\mathbb{C}^L)$ be the projection onto the subspace

$$\text{span}\{e_k : k \in J^E\}.$$

An explicit calculation of J^E is given by the following result:

Proposition 2.1.9. ([BS25, Prop. A.3.2(f)]) *Let $\sigma \in \{\pm\}$ and $E \in O_\sigma$. Then*

$$J^E = \begin{cases} \emptyset & \text{if } E \notin (-2a_1, 2a_1) \\ \{1, \dots, k\} & \text{if } E \in (-2a_k, -2a_{k+1}) \cup (2a_{k+1}, 2a_k), k = 1, \dots, L-1 \\ \{1, \dots, L\} & \text{if } E \in (-2a_L, 2a_L). \end{cases}$$

Further, for $k \in J^E$, one has

$$z_{+,k}(E) = z_{-,k}(E)^{-1}$$

and for $E \in C_+ \cap C_-$ and $k \in \{1, \dots, L\} \setminus J^E$, one has

$$z_{+,k}(E) = z_{-,k}(E).$$

With this, we can prove the following lemma:

Lemma 2.1.10. *Let $K \in \{H_0, H\}$, $\sigma, \tau \in \{\pm\}$, $E \in \overline{\mathbb{C}_\sigma} \setminus (\sigma(2A) \cup \sigma(-2A))$ and $v \in \mathbb{C}^L$. Then, the following hold:*

(a) *One has*

$$\left(m_{K,\tau}^{E,\sigma} v\right)|_{\tau\mathbb{N}} \in \ell^\infty(\tau\mathbb{N}, \mathbb{C}^L).$$

(b) *If $J^E = \emptyset$, one has*

$$\left(m_{K,\tau}^{E,\sigma} v\right)|_{\tau\mathbb{N}} \in \ell^2(\tau\mathbb{N}, \mathbb{C}^L).$$

Proof. We first notice that

$$\mathbf{1} = \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} m_{K,\tau}^{E,\sigma}(n) = \lim_{n \rightarrow \tau\infty} Z_\sigma(E)^{-\tau n} m_{K,\tau}^{E,\sigma}(n),$$

which in the case $K = H_0$ is trivial and in the case $K = H$ is precisely Theorem 2.1.6(a). This yields

$$C := \sup_{n \in \tau\mathbb{N}} \left\| Z_\sigma(E)^{-\tau n} m_{K,\tau}^{E,\sigma}(n) \right\| < \infty.$$

With this preparation, we can prove the statements:

(a) For every $n \in \tau\mathbb{N}$, we have

$$\left\| m_{K,\tau}^{E,\sigma}(n) v \right\| \leq \|Z_\sigma(E)\|^{\tau n} \cdot \left\| Z_\sigma(E)^{-\tau n} m_{K,\tau}^{E,\sigma}(n) \right\| \cdot \|v\| \leq 1 \cdot C \cdot \|v\|,$$

using Proposition 2.1.4(b) in the last step.

(b) By Proposition 2.1.4(b), using that $J^E = \emptyset$, we have

$$R := \|Z_\sigma(E)\| < 1.$$

This implies

$$\begin{aligned} \sum_{n \in \tau\mathbb{N}} \left\| m_{K,\tau}^{E,\sigma}(n) v \right\|^2 &\leq \sum_{n \in \tau\mathbb{N}} \|Z_\sigma(E)\|^{\tau 2n} \cdot \left\| Z_\sigma(E)^{-\tau n} m_{K,\tau}^{E,\sigma}(n) \right\|^2 \cdot \|v\|^2 \\ &\leq \sum_{n \in \tau\mathbb{N}} R^{\tau 2n} \cdot C^2 \|v\|^2 = \frac{R^{\tau 2}}{1 - R^{\tau 2}} \cdot C^2 \|v\|^2 < \infty. \quad \square \end{aligned}$$

2.2 Bounded generalized eigenfunctions

In this section, instead of considering all generalized eigenfunctions, we study the subspace of bounded generalized eigenfunctions since these will be used to define the scattering matrix.

Definition 2.2.1. Given $K \in \{H_0, H\}$ and an energy $E \in \mathbb{C}$, we define

$$\mathcal{B}_\pm(K, E) := \left\{ \phi \in \mathcal{G}(K, E) : \sup_{n \in \pm\mathbb{N}} |\phi(n)| < \infty \right\}$$

and

$$\mathcal{B}(K, E) := \mathcal{B}_+(K, E) \cap \mathcal{B}_-(K, E) = \left\{ \phi \in \mathcal{G}(K, E) : \sup_{n \in \mathbb{Z}} |\phi(n)| < \infty \right\}.$$

We start with the bounded eigenfunctions of the free Hamiltonian:

Lemma 2.2.2. *Let $E \in \mathbb{C}_A$. Recall notation in definition ?? . Then the following hold:*

- (a) $\mathcal{B}_\pm(H_0, E) = \text{span} \left(\left\{ \phi_\pm^{E, \sigma} e_k : k = 1, \dots, L \right\} \cup \left\{ \phi_\pm^{E, -\sigma} e_k : k \in J^E \right\} \right), E \in C_\sigma$. *In our notation, we omit the second set if J^E is the empty set. Moreover, we recall the explicit expression for J^E in Proposition 2.1.4 (f) .*
- (b) $\mathcal{B}(H_0, E) = \text{span} \{ \phi_\tau^{E, \sigma} e_k : \sigma \in \{\pm\}, k \in J^E, \tau \in \{\pm\} \}$.
- (c) $\dim \mathcal{B}(H_0, E) = 2|J^E| = \begin{cases} 0 & \text{if } E \in \mathbb{C} \setminus [-2a_1, 2a_1] \\ 2k & \text{if } E \in (-2a_k, -2a_{k+1}) \cup (2a_{k+1}, 2a_k), \\ & k = 1, \dots, L-1 \\ 2L & \text{if } E \in (-2a_L, 2a_L). \end{cases}$

Proof. (a) By Lemma ??(a), we have

$$\mathcal{G}(H_0, E) = \text{span} \left(\left\{ \phi_+^{E, \sigma} e_k : k = 1, \dots, L \right\} \cup \left\{ \phi_-^{E, \sigma} e_k : k = 1, \dots, L \right\} \right).$$

Now, for $\tau \in \{\pm\}$, one has

$$\phi_\tau^{E, \sigma}(n) e_k = Z_\sigma^{\tau n} e_k = z_{\sigma, k}^{\tau n} e_k, \quad n \in \mathbb{Z},$$

so $\phi_\tau^{E, \sigma} e_k \in \mathcal{B}_\pm(H_0, E)$, if and only if $|z_{\sigma, k}^{\pm \tau}| \leq 1$. Since, by Proposition 2.1.4(b), one has $|z_{\sigma, k}| \leq 1$, this is equivalent to $\tau = \pm$ or $|z_{\sigma, k}| = 1$, where the latter is equivalent to $k \in J^E$. This implies

$$\mathcal{B}_\pm(H_0, E) = \text{span} \left(\left\{ \phi_\pm^{E, \sigma} e_k : k = 1, \dots, L \right\} \cup \left\{ \phi_\mp^{E, \sigma} e_k : k \in J^E \right\} \right). \quad (2)$$

The statement then follows by Lemma ??(a).

(b) By Equation (2) and

$$\mathcal{B}(H_0, E) = \mathcal{B}_+(H_0, E) \cap \mathcal{B}_-(H_0, E),$$

we get

$$\mathcal{B}(H_0, E) = \text{span} \{ \phi_\tau^{E, \sigma} e_k : \tau \in \{\pm\}, k \in J^E \}.$$

The statement then follows by Equation Lemma ??(a).

(c) This follows immediately by (b) and Proposition 2.1.4(f). $\square$

The goal of this section is to show that the formula in Lemma 2.2.2(c) also holds if one replaces the free Hamiltonian H_0 with the interaction Hamiltonian H . Although this is not true for every $E \in \mathbb{C}_A$ (recall Definition ??), we will show that the set where equality does not hold is discrete. For this, we start with the following definition:

Definition 2.2.3. Let $E \in \mathbb{C}_A = C_+ \cup C_-$ (recall Definition ??) and $\tau \in \{\pm\}$. We define

$$\mathcal{E}_\tau(H, E) := \text{span} \{ u_\tau^{E, \sigma} e_k : k \in \{1, \dots, L\} \setminus J^E \}, \quad E \in C_\sigma, \sigma \in \{\pm\}.$$

(Notice that this set is well-defined, since for $E \in C_+ \cap C_-$, by Lemma ??, the right-hand side is independent of σ). Further, we set

$$D_\tau := \{E \in \mathbb{C}_A : \mathcal{E}_\tau(H, E) \cap \mathcal{B}_{-\tau}(H, E) \neq \{0\}\}$$

and define

$$D := D_+ \cup D_-.$$

We will now show that outside of the set D , the dimensions of $\mathcal{B}(H, E)$ and $\mathcal{B}(H_0, E)$ coincide. Later, we will see that D is discrete. We start by calculating the dimension of $\mathcal{B}(H, E)$ outside of D :

Theorem 2.2.4. *Let $E \in \mathbb{C}_A \setminus D$ and $\tau \in \{\pm\}$. Then the following hold:*

- (a) *For every $u \in \mathcal{E}_\tau(H, E)$, one has $\lim_{n \rightarrow \tau\infty} u(n) = 0$.*
- (b) *$\dim \mathcal{B}(H, E) = \dim \mathcal{B}(H_0, E)$.*
- (c) *$\mathcal{G}(H, E) = \mathcal{B}(H, E) \oplus \mathcal{E}_+(H, E) \oplus \mathcal{E}_-(H, E)$.*
- (d) *$\mathcal{B}_\tau(H, E) = \mathcal{B}(H, E) \oplus \mathcal{E}_\tau(H, E)$.*

Proof. (a) By Proposition 2.1.4(b), for $E \in C_\sigma$, $\sigma \in \{\pm\}$ and $k \in \{1, \dots, L\} \setminus J^E$, one has $|z_{\sigma,k}| < 1$ and therefore

$$0 = \lim_{n \rightarrow \tau\infty} z_{\sigma,k}^{\tau n} e_k = \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n) e_k = \lim_{n \rightarrow \tau\infty} u_\tau^{E,\sigma}(n) e_k,$$

using Corollary ??(b) for the last equality. The statement then follows since, by definition, the functions $\{u_\tau^{E,\sigma} e_k : k \in \{1, \dots, L\} \setminus J^E\}$ are total in $\mathcal{E}_\tau(H, E)$.

- (b) We start by showing that the following sum of vector spaces is direct:

$$\mathcal{B}(H, E) + \mathcal{E}_+(H, E) + \mathcal{E}_-(H, E).$$

So, let $\beta \in \mathcal{B}(H, E)$ and $\alpha_\pm \in \mathcal{E}_\pm(H, E)$ with $\beta + \alpha_+ + \alpha_- = 0$. Using that, by (a), one has

$$\mathcal{E}_\tau(H, E) \subseteq \mathcal{B}_\tau(H, E), \tag{3}$$

we get

$$\alpha_\pm = -\beta - \alpha_\mp \in \mathcal{B}(H, E) + \mathcal{E}_\mp(H, E) \subseteq \mathcal{B}_\mp(H, E).$$

We therefore have

$$\alpha_\pm \in \mathcal{E}_\pm(H, E) \cap \mathcal{B}_\mp(H, E) = \{0\},$$

using the assumption $E \notin D \supseteq D_\pm$ for the last equality. This yields $\alpha_+ = \alpha_- = 0$ and therefore $\beta = 0$.

Now, by Remark 2.1.2, we get

$$\begin{aligned} 2L &= \dim \mathcal{G}(H, E) \geq \dim (\mathcal{B}(H, E) \oplus \mathcal{E}_+(H, E) \oplus \mathcal{E}_-(H, E)) \\ &= \dim \mathcal{B}(H, E) + \dim \mathcal{E}_+(H, E) + \dim \mathcal{E}_-(H, E) \\ &= \dim \mathcal{B}(H, E) + 2(L - |J^E|) = 2L + \dim \mathcal{B}(H, E) - 2|J^E|, \end{aligned} \tag{4}$$

so

$$\dim \mathcal{B}(H, E) \leq 2|J^E|.$$

On the other hand, by Lemma 2.2.2(a) and Corollary ??(b), we have

$$\text{span} \left(\left\{ u_{\pm}^{E, \sigma} e_k : k = 1, \dots, L \right\} \cup \left\{ u_{\pm}^{E, -\sigma} e_k : k \in J^E \right\} \right) \subseteq \mathcal{B}_{\pm}(H, E)$$

and therefore

$$\dim \mathcal{B}_{\pm}(H, E) \geq L + |J^E|. \quad (5)$$

This yields

$$\begin{aligned} \dim \mathcal{B}(H, E) &= \dim (\mathcal{B}_+(H, E) \cap \mathcal{B}_-(H, E)) \\ &= \dim \mathcal{B}_+(H, E) + \dim \mathcal{B}_-(H, E) - \dim (\mathcal{B}_+(H, E) + \mathcal{B}_-(H, E)) \\ &\geq 2(L + |J^E|) - 2L = 2|J^E|. \end{aligned}$$

Together with (5), this implies

$$\dim \mathcal{B}(H, E) = 2|J^E| = \dim \mathcal{B}(H_0, E), \quad (6)$$

using Lemma 2.2.2(c) in the last step.

(c) By (4), we have

$$\begin{aligned} 2L &= \dim \mathcal{G}(H, E) \geq \dim (\mathcal{B}(H, E) \oplus \mathcal{E}_+(H, E) \oplus \mathcal{E}_-(H, E)) \\ &= 2L + \dim \mathcal{B}(H, E) - 2|J^E| = 2L, \end{aligned}$$

using Equation (6) in the last step. This implies

$$\dim \mathcal{G}(H, E) = 2L = \dim (\mathcal{B}(H, E) \oplus \mathcal{E}_+(H, E) \oplus \mathcal{E}_-(H, E))$$

and therefore

$$\mathcal{G}(H, E) = \mathcal{B}(H, E) \oplus \mathcal{E}_+(H, E) \oplus \mathcal{E}_-(H, E).$$

(d) By (c) and (3), we have

$$\mathcal{B}(H, E) \oplus \mathcal{E}_{\tau}(H, E) \subseteq \mathcal{B}_{\tau}(H, E).$$

Equality follows immediately by (c) and the fact that, since $E \notin D$, one has

$$\mathcal{B}_{\tau}(H, E) \cap \mathcal{E}_{-\tau}(H, E) = \{0\}. \quad \square$$

The following corollary shows that not only do the dimensions of $\mathcal{B}(H_0, E)$ and $\mathcal{B}(H, E)$ coincide, but also for every function in $\mathcal{B}(H_0, E)$, there is exactly one function in $\mathcal{B}(H, E)$ with the same asymptotic behavior:

Corollary 2.2.5. *Let $E \in \mathbb{C}_A \setminus D$ and $\tau \in \{\pm\}$. Then, the following hold:*

(a) *For every $\phi \in \mathcal{B}(H_0, E)$, there exists exactly one $u \in \mathcal{B}(H, E)$ such that*

$$\lim_{n \rightarrow \tau \infty} u(n) - \phi(n) = 0.$$

(b) *For every $u \in \mathcal{G}(H, E)$, one has $u \in \mathcal{E}_{\tau}(H, E)$, if and only if*

$$\lim_{n \rightarrow \tau \infty} u(n) = 0.$$

Proof. (a) By Lemma 2.2.2(b) and linearity, it suffices to show the statement for functions of the form $\phi = \phi_\tau^{E,\sigma} e_k$ with $\sigma \in \{\pm\}$ and $k \in J^E$. Then, for $u_\tau^{E,\sigma} e_k \in \mathcal{G}(H, E)$, by Corollary ??(b), one has

$$\lim_{n \rightarrow \tau\infty} u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n) = 0. \quad (7)$$

By Theorem 2.2.4(c), there exist unique functions $\beta \in \mathcal{B}(H, E)$ and $\alpha_\pm \in \mathcal{E}_\pm(H, E)$ such that

$$u_\tau^{E,\sigma} e_k = \beta + \alpha_+ + \alpha_-.$$

Then, by Theorem 2.2.4(d), one has

$$\alpha_{-\tau} = u_\tau^{E,\sigma} e_k - \beta - \alpha_\tau \in \mathcal{B}_\tau(H, E) + \mathcal{B}(H, E) + \mathcal{E}_\tau(H, E) = \mathcal{B}_\tau(H, E),$$

which, by $E \notin D$, implies

$$\alpha_{-\tau} \in \mathcal{E}_{-\tau}(H, E) \cap \mathcal{B}_\tau(H, E) = \{0\}.$$

Then $\alpha_{-\tau} = 0$, which yields

$$\beta = u_\tau^{E,\sigma} e_k - \alpha_\tau$$

and therefore

$$\begin{aligned} \lim_{n \rightarrow \tau\infty} \beta(n) - \phi_\tau^{E,\sigma}(n) e_k &= \lim_{n \rightarrow \tau\infty} u_\tau^{E,\sigma}(n) e_k - \alpha_\tau(n) - \phi_\tau^{E,\sigma}(n) e_k \\ &= \lim_{n \rightarrow \tau\infty} [u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n)] e_k - \alpha_\tau(n) = 0, \end{aligned}$$

using (7) and Theorem 2.2.4(a) in the last step. The existence therefore follows by setting $u = \beta$. The uniqueness follows by Theorem 2.2.4(b).

(b) Let $u \in \mathcal{G}(H, E)$ with $\lim_{n \rightarrow \tau\infty} u(n) = 0$. Then $u \in \mathcal{B}_\tau(H, E)$, so, by Theorem 2.2.4(d), there exist unique $\beta \in \mathcal{B}(H, E)$ and $\alpha_\tau \in \mathcal{E}_\tau(H, E)$, such that

$$u = \beta + \alpha_\tau.$$

Then, by Theorem 2.2.4(a), we have

$$0 = \lim_{n \rightarrow \tau\infty} u(n) - \lim_{n \rightarrow \tau\infty} \alpha_\tau(n) = \lim_{n \rightarrow \tau\infty} \beta(n).$$

By the uniqueness assertion in (a), this implies $\beta = 0$ and therefore

$$u = \alpha_\tau \in \mathcal{E}_\tau(H, E).$$

The other implication is precisely Theorem 2.2.4(a). $\square$

2.3 The Wronskian

We now want to show that the set D where our results do not apply is discrete. We will do this by linking D to the Wronskian, which is why we start this section by introducing the Wronskian for our model and showing some of its basic properties:

Definition 2.3.1. Let $N, M \in \mathbb{N}$. For matrix-valued functions $u : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N}(\mathbb{C})$ and $v : \mathbb{Z} \rightarrow \mathcal{M}_{L \times M}(\mathbb{C})$, we define

$$W(u, v)(n) := i[u(n+1)^*Av(n) - u(n)^*Av(n+1)] \in \mathcal{M}_{N \times M}(\mathbb{C}).$$

Remark 2.3.2. For every $n \in \mathbb{Z}$, one immediately sees the sesquilinearity of the map

$$(u, v) \mapsto W(u, v)(n).$$

Further, for $N_1, N_2, M_1, M_2 \in \mathbb{N}$, matrices $B \in \mathcal{M}_{N_1 \times N_2}(\mathbb{C})$, $C \in \mathcal{M}_{M_1 \times M_2}(\mathbb{C})$ and functions $u : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N_1}(\mathbb{C})$ and $v : \mathbb{Z} \rightarrow \mathcal{M}_{L \times M_1}(\mathbb{C})$, one has

$$W(uB, vC)(n) = B^*W(u, v)(n)C \quad \forall n \in \mathbb{Z}.$$

We show that, for generalized eigenfunctions, the Wronskian is independent of $n \in \mathbb{Z}$:

Proposition 2.3.3. *Let $E \in \mathbb{C}$, $N, M \in \mathbb{N}$ and $K \in \{H_0, H\}$. Further, let $v \in \mathcal{G}_N(K, E)$ and $u \in \mathcal{G}_M(K, \bar{E})$. Then $W(u, v)(n)$ is independent of $n \in \mathbb{Z}$.*

Proof. For every $n \in \mathbb{Z}$, we have

$$\begin{aligned} & -i[W(u, v)(n) - W(u, v)(n-1)] \\ &= u(n+1)^*Av(n) - u(n)^*Av(n+1) - u(n)^*Av(n-1) + u(n-1)^*Av(n) \\ &= [Au(n+1) + Au(n-1)]^*v(n) - u(n)^*[Av(n+1) + Av(n-1)] \\ &= (H_0u)(n)^*v(n) - u(n)^*(H_0v)(n). \end{aligned}$$

Since

$$\begin{aligned} & (H_0u)(n)^*v(n) - u(n)^*(H_0v)(n) \\ &= (H_0u)(n)^*v(n) - u(n)^*(H_0v)(n) + (V(n)u(n))^*v(n) - u(n)^*V(n)v(n) \\ &= ((H_0 + V)u)(n)^*v(n) - u(n)^*((H_0 + V)v)(n) \\ &= (Hu)(n)^*v(n) - u(n)^*(Hv)(n), \end{aligned}$$

this yields

$$\begin{aligned} & -i[W(u, v)(n) - W(u, v)(n-1)] \\ &= (H_0u)(n)^*v(n) - u(n)^*(H_0v)(n) \\ &= (Ku)(n)^*v(n) - u(n)^*(Kv)(n) \\ &= (\bar{E}u)(n)^*v(n) - u(n)^*(Ev)(n) \\ &= Eu(n)^*v(n) - Eu(n)^*v(n) = 0, \end{aligned}$$

so

$$W(u, v)(n) = W(u, v)(n-1) \quad \forall n \in \mathbb{Z}. \quad \square$$

In the following, we will omit the dependence on $n \in \mathbb{Z}$ and denote $W(u, v)$ if the Wronskian is independent of it as in the proposition above.

Corollary 2.3.4. Let $E \in \mathbb{C}$, $N, M \in \mathbb{N}$, $K \in \{H_0, H\}$ and $\tau \in \{\pm\}$. Further, let $v \in \mathcal{G}_N(K, E)$ and $u \in \mathcal{G}_M(K, \overline{E})$ satisfying

$$\lim_{n \rightarrow \tau\infty} u(n) = 0 \quad \text{and} \quad \limsup_{n \rightarrow \tau\infty} \|v(n)\| < \infty.$$

Then $W(u, v) = 0$.

Proof. By assumption, we have

$$C := \sup_{n \in \tau\mathbb{N}} \|v(n)\| < \infty.$$

For $n \in \tau\mathbb{N}$, this yields

$$\begin{aligned} \|W(u, v)(n)\| &= \|[u(n+1)^* Av(n) - u(n)^* Av(n+1)]\| \\ &\leq [\|u(n+1)\| + \|u(n)\|] \cdot \|A\| \cdot C \xrightarrow{n \rightarrow \tau\infty} 0, \end{aligned}$$

which implies the statement by Proposition 2.3.3. $\square$

We will now calculate the Wronskian of generalized eigenfunctions. For this, we need the following definition:

Definition 2.3.5. Let $\sigma \in \{\pm\}$ and $E \in O_\sigma$. We define

$$\omega^{E, \sigma} := -iA [Z_\sigma(E) - Z_\sigma(E)^{-1}].$$

Remark 2.3.6. By definition, for $k \in J^E$, one has

$$\begin{aligned} \omega^{E, \sigma} e_k &= -ia_k [z_{\sigma, k}(E) - z_{\sigma, k}(E)^{-1}] e_k \\ &= -ia_k [z_{\sigma, k}(E) - \overline{z_{\sigma, k}(E)}] e_k = 2a_k \operatorname{Im} z_{\sigma, k}(E) e_k. \end{aligned}$$

Since $|z_{\sigma, k}(E)| = 1$, $\operatorname{Im} z_{\sigma, k}(E) = 0$ would imply $z_{-, k}(E) = \pm 1$, which would imply

$$E = a_k [z_{-, k}(E) + z_{-, k}(E)^{-1}] = \pm 2a_k \in \sigma(\pm 2A).$$

Therefore, by Proposition 2.1.4(c), we have

$$-\sigma \langle e_k | \omega^{E, \sigma} e_k \rangle = -\sigma 2a_k \operatorname{Im} z_{\sigma, k}(E) > 0 \quad \forall k \in J^E. \quad (8)$$

Further, by Proposition 2.1.9, we get

$$\begin{aligned} \omega^{E, +} e_k &= 2a_k \operatorname{Im} z_{+, k}(E) e_k = 2a_k \operatorname{Im} z_{-, k}(E)^{-1} e_k \\ &= 2a_k \operatorname{Im} \overline{z_{-, k}(E)} e_k = -2a_k \operatorname{Im} z_{-, k}(E) e_k = -\omega^{E, -} e_k. \end{aligned} \quad (9)$$

In a similar fashion, one verifies that, for $E \in C_+ \cap C_-$ and $k \in J^E$, one has

$$\phi_\tau^{E, \sigma} e_k = \phi_{-\tau}^{E, -\sigma} e_k \quad (10)$$

and for $k \in \{1, \dots, L\} \setminus J^E$, one has

$$\phi_\tau^{E, \sigma} e_k = \phi_\tau^{E, -\sigma} e_k \quad (11)$$

(see [BS25, Lem. 2.1.8]).

Lemma 2.3.7. *Let $\sigma, \tau \in \{\pm\}$ and $E \in O_\sigma$. Then*

$$W\left(\phi_\tau^{\bar{E}, -\sigma}, \phi_\tau^{E, \sigma}\right) = 0 \quad \text{and} \quad W\left(\phi_{-\tau}^{\bar{E}, -\sigma}, \phi_\tau^{E, \sigma}\right) = \tau \omega^{E, \sigma}.$$

Proof. By Proposition 2.1.4(d), one has

$$Z_{-\sigma}(\bar{E})^* = Z_\sigma(E).$$

This yields

$$\begin{aligned} -iW(\phi_{\pm\tau}^{\bar{E}, -\sigma}, \phi_\tau^{E, \sigma}) &= \phi_{\pm\tau}^{\bar{E}, -\sigma}(n+1)^* A \phi_\tau^{E, \sigma}(n) - \phi_{\pm\tau}^{\bar{E}, -\sigma}(n)^* A \phi_\tau^{E, \sigma}(n+1) \\ &= (Z_{-\sigma}(\bar{E})^*)^{\pm\tau(n+1)} AZ_\sigma(E)^{\tau n} - (Z_{-\sigma}(\bar{E})^*)^{\pm\tau n} AZ_\sigma(E)^{\tau(n+1)} \\ &= Z_\sigma(E)^{\pm\tau(n+1)} AZ_\sigma(E)^{\tau n} - Z_\sigma(E)^{\pm\tau n} AZ_\sigma(E)^{\tau(n+1)} \\ &= A \left[Z_\sigma(E)^{\tau n \pm \tau(n+1)} - Z_\sigma(E)^{\tau(n+1) \pm \tau n} \right], \end{aligned}$$

which implies

$$\begin{aligned} -iW(\phi_\tau^{\bar{E}, -\sigma}, \phi_\tau^{E, \sigma}) &= A \left[Z_\sigma(E)^{\tau n + \tau(n+1)} - Z_\sigma(E)^{\tau(n+1) + \tau n} \right] \\ &= A \left[Z_\sigma(E)^{\tau(2n+1)} - Z_\sigma(E)^{\tau(2n+1)} \right] = 0 \end{aligned}$$

and

$$\begin{aligned} -iW(\phi_{-\tau}^{\bar{E}, -\sigma}, \phi_\tau^{E, \sigma}) &= A \left[Z_\sigma(E)^{\tau n - \tau(n+1)} - Z_\sigma(E)^{\tau(n+1) - \tau n} \right] \\ &= A \left[Z_\sigma(E)^{-\tau 1} - Z_\sigma(E)^{\tau 1} \right] = -i\tau \omega^{E, \sigma}. \quad \square \end{aligned}$$

We will now prove some technical lemmas that we will need later to prove that the set D is discrete:

Lemma 2.3.8. *Let $\sigma \in \{\pm\}$. Then*

$$\lim_{E \in C_\sigma, |E| \rightarrow \infty} E^{-1} \omega^{E, \sigma} = i\mathbf{1}.$$

Proof. One has

$$\begin{aligned} E^{-1} \omega^{E, \sigma} &= iE^{-1} A \left[Z_\sigma(E)^{-1} - Z_\sigma(E) \right] \\ &= iE^{-1} \left[A \left(Z_\sigma(E)^{-1} + Z_\sigma(E) \right) - 2AZ_\sigma(E) \right] \\ &= iE^{-1} [E - 2AZ_\sigma(E)] \\ &= i\mathbf{1} - iE^{-1} 2AZ_\sigma(E) \xrightarrow{|E| \rightarrow \infty} i\mathbf{1}, \end{aligned}$$

using in the last step that, by Proposition 2.1.4(b), one has $\|Z_\sigma(E)\| \leq 1$ and therefore

$$\|iE^{-1} 2AZ_\sigma(E)\| = |E|^{-1} 2\|A\| \cdot \|Z_\sigma(E)\| \leq |E|^{-1} 2\|A\| \xrightarrow{|E| \rightarrow \infty} 0. \quad \square$$

Lemma 2.3.9. *Let $\sigma, \tau \in \{\pm\}$, $N \in \mathbb{N}$ and $B, C \in \mathcal{M}_{L \times N}(\mathbb{C})$ with $B^*C \in \text{GL}_N(\mathbb{C})$. Then, the set of energies $E \in C_\sigma^\circ$ (or $E \in O_\sigma$ if the strict growth condition holds), such that*

$$W(u_{-\tau}^{\bar{E}, -\sigma} B, u_\tau^{E, \sigma} C) \in \mathcal{M}_N(\mathbb{C})$$

is not invertible, is discrete (here the symbol Θ° stands for the interior of the set Θ).

Proof. The function

$$E \mapsto \det \left(-iW(u_{-\tau}^{\bar{E}, -\sigma} B, u_{\tau}^{E, \sigma} C) \right)$$

is holomorphic on C_{σ}° or O_{σ} respectively. It therefore suffices to show that this function is not constantly zero, since, by the identity theorem for holomorphic functions, the zeros of any non-zero holomorphic function are discrete. If constantly zero, we would have

$$\begin{aligned} 0 &= \lim_{|E| \rightarrow \infty} E^{-N} \det \left(-iW(u_{-\tau}^{\bar{E}, -\sigma} B, u_{\tau}^{E, \sigma} C) \right) = \lim_{|E| \rightarrow \infty} \det \left(-iE^{-1}W(u_{-\tau}^{\bar{E}, -\sigma} B, u_{\tau}^{E, \sigma} C) \right) \\ &= \lim_{|E| \rightarrow \infty} \det \left(-iE^{-1}W(\phi_{-\tau}^{\bar{E}, -\sigma} B, \phi_{\tau}^{E, \sigma} C) \right), \end{aligned}$$

using Theorem 2.1.6(f) in the last step. This contradicts that, by Lemma 2.3.7, we have

$$-iE^{-1}W(\phi_{-\tau}^{\bar{E}, -\sigma} B, \phi_{\tau}^{E, \sigma} C) = -iB^*E^{-1}\tau\omega^{E, \sigma}C \xrightarrow{|E| \rightarrow \infty} \tau B^*C$$

and therefore

$$\lim_{|E| \rightarrow \infty} \det \left(-iE^{-1}W(\phi_{-\tau}^{\bar{E}, -\sigma} B, \phi_{\tau}^{E, \sigma} C) \right) = \det(\tau B^*C) \neq 0. \quad \square$$

We now want to link the Wronskian, appearing in the above lemma, to the set D . For this, we need the following definition:

Definition 2.3.10. Given $M \in \{1, \dots, L\}$ and a subset $I = \{i_1, \dots, i_M\} \subseteq \{1, \dots, L\}$ with $i_1 < \dots < i_M$, we define

$$P_I \in \mathcal{M}_{L \times M}(\mathbb{C})^{\mathbb{Z}}$$

by

$$(P_I)_{jk} = \delta_{\cancel{i_k}k} \delta_{j, i_k}, \quad j \in \{1, \dots, L\}, \quad k \in \{1, \dots, M\}$$

.

Lemma 2.3.11. Let $\sigma, \tau \in \{\pm\}$, $E \in D_{\tau} \cap C_{\sigma}$, and set $P := P_{\{1, \dots, L\} \setminus J^E}$. Then the matrix

$$W(u_{-\tau}^{\bar{E}, -\sigma} P, u_{\tau}^{E, \sigma} P) \in \mathcal{M}_{L-|J^E|}(\mathbb{C})$$

is not invertible.

Proof. By definition of D_{τ} (Definition 2.2.3), there exists $v \in \mathbb{C}^{L-|J^E|} \setminus \{0\}$, such that

$$u_{\tau}^{E, \sigma} P v \in \mathcal{E}_{\tau}(H, E) \cap \mathcal{B}_{-\tau}(H, E).$$

This implies that

$$\limsup_{n \rightarrow -\tau\infty} \|u_{\tau}^{E, \sigma}(n) P v\| < \infty.$$

On the other hand, by Theorem 2.2.4(a), we have

$$\lim_{n \rightarrow -\tau\infty} \|u_{-\tau}^{\bar{E}, -\sigma} P\| = 0.$$

Therefore, by Corollary 2.3.4, we have

$$0 = W(u_{-\tau}^{\bar{E}, -\sigma} P, u_{\tau}^{E, \sigma} P v) = W(u_{-\tau}^{\bar{E}, -\sigma} P, u_{\tau}^{E, \sigma} P) v,$$

wherefore $W(u_{-\tau}^{\bar{E}, -\sigma} P, u_{\tau}^{E, \sigma} P)$ is not invertible, since it has non-trivial kernel. $\square$

Finally, the following proposition shows that D is discrete, and therefore the statements of the previous section hold for almost every complex energy:

Proposition 2.3.12. *The set D is a discrete subset of $\mathbb{C}_A \setminus (-2a_L, 2a_L)$.*

Proof. For every $E \in (-2a_L, 2a_L)$, by Proposition 2.1.4(f), one has $J^E = \{1, \dots, L\}$ and therefore $\mathcal{E}_\tau(H, E) = \{0\}$, $\tau \in \{\pm\}$, which implies $E \notin D$. We therefore have

$$D \subseteq \mathbb{C}_A \setminus (-2a_L, 2a_L).$$

In the case $\tau_A = 1$, we have $(-2a_L, 2a_L) = (-2a_1, 2a_1)$ and therefore

$$\mathbb{C}_A \setminus (-2a_L, 2a_L) = \mathbb{C}_A \setminus (-2a_1, 2a_1) = C_+^\circ \cup C_-^\circ.$$

Now, for $k = 1, \dots, L$, we have

$$(P_{\{k, \dots, L\}})^* P_{\{k, \dots, L\}} = \mathbf{1}_{L-k+1} \in \text{GL}_{L-k+1}(\mathbb{C}).$$

Therefore, for $\sigma, \tau \in \pm$, by Lemma 2.3.9, the set

$$M_{\tau, k}^\sigma := \left\{ E \in C_\sigma^\circ : W \left(u_{-\tau}^{\bar{E}, -\sigma} P_{\{k, \dots, L\}}, u_\tau^{E, \sigma} P_{\{k, \dots, L\}} \right) \notin \text{GL}_{L-k+1}(\mathbb{C}) \right\} \subseteq C_\sigma^\circ$$

is discrete. By Lemma 2.3.11, we have

$$D \subseteq \bigcup_{\substack{\sigma, \tau \in \{\pm\} \\ k=1, \dots, L}} M_{\tau, k}^\sigma,$$

which implies that also D is discrete in $C_+^\circ \cup C_-^\circ = \mathbb{C}_A \setminus (-2a_L, 2a_L)$.

If $\tau_A > 1$, the strict growth condition is fulfilled (this saw not clearly stated in the hypothesis), and we have

$$\mathbb{C}_A = O_+ \cup O_-.$$

By replacing $C_\pm^\circ$ with $O_\pm$ in the argument above, one sees that D is discrete in the set $O_+ \cup O_- = \mathbb{C}_A \supseteq \mathbb{C}_A \setminus (-2a_L, 2a_L)$. $\square$

3 Analysis of the resolvent

3.1 The Green's function

In this section, we use our previous results about generalized eigenfunctions to calculate the Green's function for the free and interacting Haltonian and prove some of their properties. We define them as follows:

Definition 3.1.1. Let $K \in \{H_0, H\}$, $\sigma, \tau \in \{\pm\}$ and $E \in O_\alpha^\sigma \setminus \mathfrak{D}$. We define

$$W_{K, \tau}^{E, \sigma} := W \left(m_{K, -\tau}^{\bar{E}, -\sigma}, m_{K, \tau}^{E, \sigma} \right)$$

and set

$$G_K^{E, \sigma}(n, j) = \begin{cases} -m_{K, +}^{E, \sigma}(n) i \left(W_{K, +}^{E, \sigma} \right)^{-1} m_{K, -}^{\bar{E}, -\sigma}(j)^* & \text{if } n > j \\ m_{K, -}^{E, \sigma}(n) i \left(W_{K, -}^{E, \sigma} \right)^{-1} m_{K, +}^{\bar{E}, -\sigma}(j)^* & \text{if } n \leq j. \end{cases}$$

We now want to show that $G^{E,\sigma}$ are actually the Green's functions for our Hamiltonians. For this, we need the following lemma, showing that in the definition of $G_K^{E,\sigma}$, instead of considering the cases $n > j$ and $n \leq j$, one might as well have chosen $n \geq j$ and $n < j$, or in other words: For $n = j$ the two terms in the definition of $G_K^{E,\sigma}$ coincide.

Lemma 3.1.2. *Let $K \in \{H_0, H\}$, $\sigma \in \{\pm\}$ and $E \in O_\alpha^\sigma \setminus \mathfrak{D}$. Then*

$$Am_{K,+}^{E,\sigma}(j)i\left(W_{K,+}^{E,\sigma}\right)^{-1}m_{K,-}^{\bar{E},-\sigma}(j)^* + Am_{K,-}^{E,\sigma}(j)i\left(W_{K,-}^{E,\sigma}\right)^{-1}m_{K,+}^{\bar{E},-\sigma}(j)^* = 0$$

and

$$-Am_{K,+}^{E,\sigma}(j+1)i\left(W_{K,+}^{E,\sigma}\right)^{-1}m_{K,-}^{\bar{E},-\sigma}(j)^* - Am_{K,-}^{E,\sigma}(j+1)i\left(W_{K,-}^{E,\sigma}\right)^{-1}m_{K,+}^{\bar{E},-\sigma}(j)^* = \mathbf{1}_L$$

Proof. We define

$$T^{E,\sigma} := \begin{pmatrix} m_{K,+}^{E,\sigma}(j) & -m_{K,-}^{E,\sigma}(j) \\ m_{K,+}^{E,\sigma}(j+1) & -m_{K,-}^{E,\sigma}(j+1) \end{pmatrix} \in \mathcal{M}_{2L \times 2L}(\mathbb{C}).$$

We then have

$$\begin{aligned} & \left(T^{\bar{E},-\sigma}\right)^* \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} T^{E,\sigma} \\ &= \begin{pmatrix} m_{K,+}^{\bar{E},-\sigma}(j)^* & m_{K,+}^{\bar{E},-\sigma}(j+1)^* \\ -m_{K,-}^{\bar{E},-\sigma}(j)^* & -m_{K,-}^{\bar{E},-\sigma}(j+1)^* \end{pmatrix} \\ & \quad \cdot \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} \begin{pmatrix} m_{K,+}^{E,\sigma}(j) & -m_{K,-}^{E,\sigma}(j) \\ m_{K,+}^{E,\sigma}(j+1) & -m_{K,-}^{E,\sigma}(j+1) \end{pmatrix} \\ &= \begin{pmatrix} m_{K,+}^{\bar{E},-\sigma}(j)^* & m_{K,+}^{\bar{E},-\sigma}(j+1)^* \\ -m_{K,-}^{\bar{E},-\sigma}(j)^* & -m_{K,-}^{\bar{E},-\sigma}(j+1)^* \end{pmatrix} \begin{pmatrix} Am_{K,+}^{E,\sigma}(j+1) & -Am_{K,-}^{E,\sigma}(j+1) \\ -Am_{K,+}^{E,\sigma}(j) & Am_{K,-}^{E,\sigma}(j) \end{pmatrix} \\ &= \begin{pmatrix} iW\left(m_{K,+}^{\bar{E},-\sigma}, m_{K,+}^{E,\sigma}\right) & -iW\left(m_{K,+}^{\bar{E},-\sigma}, m_{K,-}^{E,\sigma}\right) \\ -iW\left(m_{K,-}^{\bar{E},-\sigma}, m_{K,+}^{E,\sigma}\right) & iW\left(m_{K,-}^{\bar{E},-\sigma}, m_{K,-}^{E,\sigma}\right) \end{pmatrix} \\ &= \begin{pmatrix} 0 & -iW_{K,-}^{E,\sigma} \\ -iW_{K,+}^{E,\sigma} & 0 \end{pmatrix} \end{aligned} \tag{12}$$

using in the last step that, by Theorem 2.1.6(b) and Lemma 2.3.7, we have

$$W\left(m_{K,\pm}^{\bar{E},-\sigma}, m_{K,\pm}^{E,\sigma}\right) = W\left(\phi_{\pm}^{\bar{E},-\sigma}, \phi_{\pm}^{E,\sigma}\right) = 0.$$

Since the right-hand side of Equation (12) is invertible, this implies the invertibility of the

matrices $T^{\bar{E}, -\sigma}$ and $T^{E, \sigma}$. Further, Equation (12) yields

$$\begin{aligned}
& (T^{\bar{E}, -\sigma})^* \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} T^{E, \sigma} \begin{pmatrix} -i \left(W_{K,+}^{E, \sigma} \right)^{-1} m_{K,-}^{\bar{E}, -\sigma}(j)^* \\ i \left(W_{K,-}^{E, \sigma} \right)^{-1} m_{K,+}^{\bar{E}, -\sigma}(j)^* \end{pmatrix} \\
&= \begin{pmatrix} m_{K,+}^{\bar{E}, -\sigma}(j)^* \\ -m_{K,-}^{\bar{E}, -\sigma}(j)^* \end{pmatrix} = \begin{pmatrix} m_{K,+}^{\bar{E}, -\sigma}(j)^* & m_{K,+}^{\bar{E}, -\sigma}(j+1)^* \\ -m_{K,-}^{\bar{E}, -\sigma}(j)^* & -m_{K,-}^{\bar{E}, -\sigma}(j+1)^* \end{pmatrix} \begin{pmatrix} \mathbf{1}_L \\ 0 \end{pmatrix} \\
&= (T^{\bar{E}, -\sigma})^* \begin{pmatrix} \mathbf{1}_L \\ 0 \end{pmatrix},
\end{aligned}$$

which, together with the already established invertability of $T^{\bar{E}, -\sigma}$, yields

$$\begin{aligned}
\begin{pmatrix} \mathbf{1}_L \\ 0 \end{pmatrix} &= \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} T^{E, \sigma} \begin{pmatrix} -i \left(W_{K,+}^{E, \sigma} \right)^{-1} m_{K,-}^{\bar{E}, -\sigma}(j)^* \\ i \left(W_{K,-}^{E, \sigma} \right)^{-1} m_{K,+}^{\bar{E}, -\sigma}(j)^* \end{pmatrix} \\
&= \begin{pmatrix} Am_{K,+}^{E, \sigma}(j+1) & -Am_{K,-}^{E, \sigma}(j+1) \\ -Am_{K,+}^{E, \sigma}(j) & Am_{K,-}^{E, \sigma}(j) \end{pmatrix} \begin{pmatrix} -i \left(W_{K,+}^{E, \sigma} \right)^{-1} m_{K,-}^{\bar{E}, -\sigma}(j)^* \\ i \left(W_{K,-}^{E, \sigma} \right)^{-1} m_{K,+}^{\bar{E}, -\sigma}(j)^* \end{pmatrix} \\
&= \begin{pmatrix} -Am_{K,+}^{E, \sigma}(j+1)i \left(W_{K,+}^{E, \sigma} \right)^{-1} m_{K,-}^{\bar{E}, -\sigma}(j)^* - Am_{K,-}^{E, \sigma}(j+1)i \left(W_{K,-}^{E, \sigma} \right)^{-1} m_{K,+}^{\bar{E}, -\sigma}(j)^* \\ Am_{K,+}^{E, \sigma}(j)i \left(W_{K,+}^{E, \sigma} \right)^{-1} m_{K,-}^{\bar{E}, -\sigma}(j)^* + Am_{K,-}^{E, \sigma}(j)i \left(W_{K,-}^{E, \sigma} \right)^{-1} m_{K,+}^{\bar{E}, -\sigma}(j)^* \end{pmatrix},
\end{aligned}$$

giving us the two desired equations. $\square$

With this preparation, we can prove that $G^{E, \sigma}$ are actually the Green's functions for our Hamiltonians:

Theorem 3.1.3. *Let $K \in \{H_0, H\}$, $\sigma \in \{\pm\}$ and $E \in O_\alpha^\sigma \setminus \mathfrak{D}$. Then, the following hold:*

(a) *For every $j \in \mathbb{Z}$, one has*

$$(K - E)G_K^{E, \sigma}(\cdot, j) = \delta_j \cdot \mathbf{1}_L.$$

(b) *If $E \in \overline{\mathbb{C}_\sigma}$, for every $j \in \mathbb{Z}$, one has*

$$G_K^{E, \sigma}(\cdot, j) \in \ell^\infty(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C})).$$

(c) *If $E \notin [-2a_1, 2a_1]$, for every $j \in \mathbb{Z}$, one has*

$$G_K^{E, \sigma}(\cdot, j) \in \ell^2(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C})).$$

(d) *For every $n, j \in \mathbb{Z}$, one has*

$$G_K^{E, \sigma}(n, j) = \left(G_K^{\bar{E}, -\sigma}(j, n) \right)^*.$$

(e) *If $E \in \overline{\mathbb{C}_\sigma}$, for every $n \in \mathbb{Z}$, one has*

$$G_K^{E, \sigma}(n, \cdot) \in \ell^\infty(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C})).$$

(f) If $E \notin [-2a_1, 2a_1]$, for every $n \in \mathbb{Z}$, one has

$$G_K^{E,\sigma}(n, \cdot) \in \ell^2(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C})).$$

(g) If $E \notin \sigma(K)$, for every $f \in \ell^2(\mathbb{Z}, \mathbb{C}^L)$, one has

$$(K - E)^{-1}f = \sum_{j \in \mathbb{Z}} G_K^{E,\sigma}(\cdot, j)f(j).$$

(h) For $K = H_0$, for $n, j \in \mathbb{Z}$, one has

$$G_{H_0}^{E,\sigma}(n, j) = (i\omega^{E,\sigma})^{-1} Z_\sigma(E)^{|n-j|}.$$

Proof. (a) Using that $m_{K,+}^{E,\sigma}, m_{K,-}^{E,\sigma} \in \mathcal{G}_L(K, E)$, we immediately get

$$\left((K - E)G_K^{E,\sigma}(\cdot, j) \right)(n) = 0 \quad \forall n \notin \{j, j+1\}.$$

Further, by Lemma 3.1.2, one has

$$\begin{aligned} \left((K - E)G_K^{E,\sigma}(\cdot, j) \right)(j+1) &= - \left((K - E)m_{K,+}^{E,\sigma} \right)(j+1) \cdot i \left(W_{K,+}^{E,\sigma} \right)^{-1} m_{K,-}^{\bar{E},-\sigma}(j)^* \\ &\quad + Am_{K,+}^{E,\sigma}(j) i \left(W_{K,+}^{E,\sigma} \right)^{-1} m_{K,-}^{\bar{E},-\sigma}(j)^* \\ &\quad + Am_{K,-}^{E,\sigma}(j) i \left(W_{K,-}^{E,\sigma} \right)^{-1} m_{K,+}^{\bar{E},-\sigma}(j)^* \\ &= 0 \cdot i \left(W_{K,+}^{E,\sigma} \right)^{-1} m_{K,-}^{\bar{E},-\sigma}(j)^* + 0 = 0 \end{aligned}$$

and

$$\begin{aligned} \left((K - E)G_K^{E,\sigma}(\cdot, j) \right)(j) &= \left((K - E)m_{K,-}^{E,\sigma} \right)(j) \cdot i \left(W_{K,-}^{E,\sigma} \right)^{-1} m_{K,+}^{\bar{E},-\sigma}(j)^* \\ &\quad - Am_{K,-}^{E,\sigma}(j+1) i \left(W_{K,-}^{E,\sigma} \right)^{-1} m_{K,+}^{\bar{E},-\sigma}(j)^* \\ &\quad - Am_{K,+}^{E,\sigma}(j+1) i \left(W_{K,+}^{E,\sigma} \right)^{-1} m_{K,-}^{\bar{E},-\sigma}(j)^* \\ &= 0 \cdot i \left(W_{K,-}^{E,\sigma} \right)^{-1} m_{K,+}^{\bar{E},-\sigma}(j)^* + \mathbf{1}_L = \mathbf{1}_L. \end{aligned}$$

(b) This follows immediately by Lemma 2.1.10(a).

(c) Since $E \notin [-2a_1, 2a_1]$, by Proposition 2.1.9, we have $J^E = \emptyset$. The statement then follows by Lemma 2.1.10(b).

(d) Since the functions

$$E \mapsto G_K^{E,\sigma}(n, j), \quad n, j \in \mathbb{Z}, \sigma \in \{\pm\}$$

are continuous by Theorem 2.1.6(a) and Proposition ??, it suffices to show the statement for $E \notin [-2a_1, 2a_1]$. Then, by (c), for $v, w \in \mathbb{C}^L$, we have

$$G_K^{E,\sigma}(\cdot, j)v, G_K^{\bar{E},-\sigma}(\cdot, j)w \in \ell^2(\mathbb{Z}, \mathbb{C}^L) = \mathcal{H}.$$

Then, by (a), we have

$$\begin{aligned}
\left\langle w \left| G_K^{E,\sigma}(n, j) v \right. \right\rangle_{\mathbb{C}^L} &= \left\langle \delta_n w \left| G_K^{E,\sigma}(\cdot, j) v \right. \right\rangle_{\mathcal{H}} = \left\langle (K - \bar{E}) G_K^{\bar{E}, -\sigma}(\cdot, n) w \left| G_K^{E,\sigma}(\cdot, j) v \right. \right\rangle_{\mathcal{H}} \\
&= \left\langle G_K^{\bar{E}, -\sigma}(\cdot, n) w \left| (K - E) G_K^{E,\sigma}(\cdot, j) v \right. \right\rangle_{\mathcal{H}} = \left\langle G_K^{\bar{E}, -\sigma}(\cdot, n) w \left| \delta_j v \right. \right\rangle_{\mathcal{H}} \\
&= \left\langle G_K^{\bar{E}, -\sigma}(j, n) w \left| v \right. \right\rangle_{\mathbb{C}^L} = \left\langle w \left| \left(G_K^{\bar{E}, -\sigma}(j, n) \right)^* v \right. \right\rangle_{\mathbb{C}^L}.
\end{aligned}$$

(e) This follows immediately by (b) and (d).

(f) This follows immediately by (c) and (d).

(g) The sum $\sum_{j \in \mathbb{Z}} G_K^{E,\sigma}(\cdot, j) f(j)$ converges absolutely by (f). Then, by (a), we have

$$(K - E) \sum_{j \in \mathbb{Z}} G_K^{E,\sigma}(\cdot, j) f(j) = \sum_{j \in \mathbb{Z}} (K - E) G_K^{E,\sigma}(\cdot, j) f(j) = \sum_{j \in \mathbb{Z}} \delta_j f(j) = f.$$

(h) By Lemma 2.3.7 and Proposition 2.1.4(d), we get

$$\begin{aligned}
m_{H_0, \pm}^{E, \sigma}(n) i \left(W \left(m_{H_0, \mp}^{\bar{E}, -\sigma}, m_{H_0, \pm}^{E, \sigma} \right) \right)^{-1} m_{H_0, \mp}^{\bar{E}, -\sigma}(j)^* \\
&= \phi_{\pm}^{E, \sigma}(n) i \left(W \left(\phi_{\mp}^{\bar{E}, -\sigma}, \phi_{\pm}^{E, \sigma} \right) \right)^{-1} \phi_{\mp}^{\bar{E}, -\sigma}(j)^* \\
&= Z_{\sigma}(E)^{\pm n} i \left(\pm \omega^{E, \sigma} \right)^{-1} \left(Z_{-\sigma}(\bar{E})^{\mp j} \right)^* \\
&= \mp \left(i \omega^{E, \sigma} \right)^{-1} Z_{\sigma}(E)^{\pm n} Z_{\sigma}(E)^{\mp j} \\
&= \mp \left(i \omega^{E, \sigma} \right)^{-1} Z_{\sigma}(E)^{\pm(n-j)}.
\end{aligned}$$

Plugging this into (??) yields the desired formula for $G_{H_0}^{E, \sigma}(n, j)$. $\square$

We finish this section by proving some properties of the Green's function $G_K^{E, \sigma}$:

Lemma 3.1.4. *Let $K \in \{H_0, H\}$, $\sigma \in \{\pm\}$ and $n, j \in \mathbb{Z}$. Then, the following hold:*

(a) *The function*

$$\overline{\mathbb{C}_{\sigma}} \setminus \mathfrak{D} \ni E \mapsto G_K^{E, \sigma}(n, j) \in \mathcal{M}_{L \times L}(\mathbb{C})$$

is continuous.

(b) *For every compact subset $F \subseteq \overline{\mathbb{C}_{\sigma}} \setminus \mathfrak{D}$, one has*

$$\sup_{E \in F} \left\| G_K^{E, \sigma}(\cdot, j) \right\|_{\infty} < \infty \quad \text{and} \quad \sup_{E \in F} \left\| G_K^{E, \sigma}(n, \cdot) \right\|_{\infty} < \infty.$$

(c) *The functions*

$$\overline{\mathbb{C}_{\sigma}} \setminus \mathfrak{D} \ni E \mapsto V G_K^{E, \sigma}(\cdot, j) \in \ell^1(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C}))$$

and

$$\overline{\mathbb{C}_{\sigma}} \setminus \mathfrak{D} \ni E \mapsto G_K^{E, \sigma}(n, \cdot) V \in \ell^1(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C}))$$

and

$$\overline{\mathbb{C}_{\sigma}} \setminus \mathfrak{D} \ni E \mapsto G_K^{E, \sigma}(n, \cdot) V G_K^{E, \sigma}(\cdot, j) \in \ell^1(\mathbb{Z}, \mathcal{M}_{L \times L}(\mathbb{C}))$$

are continuous.

Proof. (a) This follows immediately by Theorem 2.1.6(a) and Theorem 3.1.3(h).

(b) The first statement follows immediately by Theorem 2.1.6(c) and the continuity of the functions

$$F \ni E \mapsto \left(W_{K,\pm}^{E,\sigma}(j) \right)^{-1}$$

due to Proposition ???. The second statement then follows by Theorem 3.1.3(d).

(c) We show that the maps are sequentially continuous. So let $(x_k)_{k \in \mathbb{N}}$ be a sequence in $\overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$ with limit $x \in \overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$. By (a), for every $l \in \mathbb{Z}$, one has

$$\lim_{k \rightarrow \infty} G_K^{x_k, \sigma}(l, j) = G_K^{x, \sigma}(l, j).$$

Now, the set

$$F := \{x_k : k \in \mathbb{N}\} \cup \{x\} \subseteq \overline{\mathbb{C}_\sigma} \setminus \mathfrak{D}$$

is compact, and therefore, by (b), we have

$$\sup_{E \in F} \left\| G_K^{E, \sigma}(\cdot, j) \right\|_\infty < \infty.$$

Using the Dominated Convergence Theorem, this yields

$$\lim_{k \rightarrow \infty} \sum_{l \in \mathbb{Z}} \|V(l) [G_K^{x_k, \sigma}(l, j) - G_K^{x, \sigma}(l, j)]\| = 0$$

and

$$\lim_{k \rightarrow \infty} \sum_{l \in \mathbb{Z}} \|[G_K^{x_k, \sigma}(n, l) - G_K^{x, \sigma}(n, l)] V(l)\| = 0$$

and

$$\lim_{k \rightarrow \infty} \sum_{l \in \mathbb{Z}} \|G_K^{x_k, \sigma}(n, l) V(l) G_K^{x_k, \sigma}(l, j) - G_K^{x, \sigma}(n, l) V(l) G_K^{x, \sigma}(l, j)\| = 0. \quad \square$$

3.2 Limiting absorption principle

In this section, we will analyze the behavior of the resolvent of the operators H_0 and H when approaching the spectrum. Given a bounded function $b \in \ell^\infty(\mathbb{Z}, \mathbb{C}^L)$, using that

$$\sum_{n \in \mathbb{Z}} \|V(n)\| \leq \|V(0)\| + \sum_{n \in \mathbb{Z}} \|V(n)\| |n| \tau_A^{|n|} < \infty, \quad (13)$$

we have $Vb \in \ell^1(\mathbb{Z}, \mathbb{C}^L) \subseteq \ell^2(\mathbb{Z}, \mathbb{C}^L)$. Then, given $K \in \{H_0, H\}$ and an energy $E \in \mathbb{R}$, one can ask if, in some sense, the limits

$$\lim_{\varepsilon \downarrow 0} (K - E \mp i\varepsilon)^{-1} (Vb)$$

exist. In general, these limits don't exist in $\ell^2(\mathbb{Z}, \mathbb{C}^L)$, but, as we will see, pointwise limits exist in $\ell^\infty(\mathbb{Z}, \mathbb{C}^L)$ as long as b is a bounded generalized eigenfunction.

Theorem 3.2.1. *Let $E \in [-2a_1, 2a_1] \setminus \mathfrak{D}$ and $\sigma \in \{\pm\}$. Then, the following hold:*

(a) For every $b \in \mathcal{B}(H, E)$ and every $n \in \mathbb{Z}$, the limit

$$(\Phi_\sigma^E b)(n) := \lim_{\varepsilon \downarrow 0} ((H_0 - E - \sigma i\varepsilon)^{-1}(H - E - \sigma i\varepsilon)b)(n)$$

exists and defines a function $\Phi_\sigma^E b \in \mathcal{B}(H_0, E)$. Concretely, one has

$$\begin{aligned} \Phi_\sigma^E b &= m_{H_0, -}^{E, -\sigma} \cdot \left((W_{H_0, +}^{E, \sigma})^{-1} \right)^* W(m_{H, +}^{E, \sigma}, b) \\ &\quad + m_{H_0, +}^{E, -\sigma} \cdot \left((W_{H_0, -}^{E, \sigma})^{-1} \right)^* W(m_{H, -}^{E, \sigma}, b). \end{aligned} \quad (14)$$

(b) For every $b \in \mathcal{B}(H_0, E)$ and every $n \in \mathbb{Z}$, the limit

$$(\Psi_\sigma^E b)(n) := \lim_{\varepsilon \downarrow 0} ((H - E - \sigma i\varepsilon)^{-1}(H_0 - E - \sigma i\varepsilon)b)(n)$$

exists and defines a function $\Psi_\sigma^E b \in \mathcal{G}(H, E)$. Concretely, one has

$$\begin{aligned} \Psi_\sigma^E b &= m_{H, -}^{E, -\sigma} \cdot \left((W_{H, +}^{E, \sigma})^{-1} \right)^* W(m_{H_0, +}^{E, \sigma}, b) \\ &\quad + m_{H, +}^{E, -\sigma} \cdot \left((W_{H, -}^{E, \sigma})^{-1} \right)^* W(m_{H_0, -}^{E, \sigma}, b). \end{aligned} \quad (15)$$

(c) For every $b \in \mathcal{B}(H, E)$, one has $(\Psi_\sigma^E \circ \Phi_\sigma^E)b = b$.

(d) If $E \in \mathfrak{D}'$, one has $\Psi_\sigma^E(\mathcal{B}(H_0, E)) \subseteq \mathcal{B}(H, E)$ and the map

$$\Psi_\sigma^E : \mathcal{B}(H_0, E) \rightarrow \mathcal{B}(H, E)$$

is an isomorphism.

Proof. (a) We first notice that, by Theorem 3.1.3(g) and Lemma 3.1.4(c), we have

$$\begin{aligned} (\Phi_\sigma^E b)(n) &= \lim_{\varepsilon \downarrow 0} ((H_0 - E - \sigma i\varepsilon)^{-1}(H - E - \sigma i\varepsilon)b)(n) \\ &= b(n) + \lim_{\varepsilon \downarrow 0} ((H_0 - E - \sigma i\varepsilon)^{-1}Vb)(n) \\ &= b(n) + \lim_{\varepsilon \downarrow 0} \sum_{j \in \mathbb{Z}} G_{H_0}^{E + \sigma i\varepsilon, \sigma}(n, j)V(j)b(j) \\ &= b(n) + \sum_{j \in \mathbb{Z}} G_{H_0}^{E, \sigma}(n, j)V(j)b(j). \end{aligned} \quad (16)$$

Since $b \in \mathcal{B}(H, E)$, we have

$$Vb = (H - H_0)b = (H - E)b - (H_0 - E)b = (E - H_0)b.$$

For $k < n < l$, this yields

$$\begin{aligned}
& \sum_{j=k}^l G_{H_0}^{E,\sigma}(n, j) V(j) b(j) \\
&= \sum_{j=k}^l G_{H_0}^{E,\sigma}(n, j) [Eb(j) - Ab(j+1) - Ab(j-1)] \\
&= \sum_{j=k}^l G_{H_0}^{E,\sigma}(n, j) Eb(j) - \sum_{j=k+1}^{l+1} G_{H_0}^{E,\sigma}(n, j-1) Ab(j) - \sum_{j=k-1}^{l-1} G_{H_0}^{E,\sigma}(n, j+1) Ab(j) \\
&= \sum_{j=k}^l \left[G_{H_0}^{E,\sigma}(n, j) E - G_{H_0}^{E,\sigma}(n, j-1) A - G_{H_0}^{E,\sigma}(n, j+1) A \right] b(j) \\
&\quad - G_{H_0}^{E,\sigma}(n, l) Ab(l+1) + G_{H_0}^{E,\sigma}(n, l+1) Ab(l) \\
&\quad + G_{H_0}^{E,\sigma}(n, k-1) Ab(k) - G_{H_0}^{E,\sigma}(n, k) Ab(k-1) \\
&= \sum_{j=k}^l \left[EG_{H_0}^{E,-\sigma}(j, n) - AG_{H_0}^{E,-\sigma}(j-1, n) - AG_{H_0}^{E,-\sigma}(j+1, n) \right]^* b(j) \\
&\quad - G_{H_0}^{E,-\sigma}(l, n)^* Ab(l+1) + G_{H_0}^{E,-\sigma}(l+1, n)^* Ab(l) \\
&\quad + G_{H_0}^{E,-\sigma}(k-1, n)^* Ab(k) - G_{H_0}^{E,-\sigma}(k, n)^* Ab(k-1) \\
&= \sum_{j=k}^l \left[(E - H_0) G_{H_0}^{E,-\sigma}(\cdot, n)(j) \right]^* b(j) \\
&\quad - iW \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (l) + iW \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (k-1),
\end{aligned}$$

using Theorem 3.1.3(d) in the penultimate step. Using that, by Theorem 3.1.3(a), we have

$$\sum_{j=k}^l \left[(E - H_0) G_{H_0}^{E,-\sigma}(\cdot, n)(j) \right]^* b(j) = \sum_{j=k}^l [-\delta_j(n) \cdot \mathbf{1}_L]^* b(j) = -b(n),$$

this yields

$$\begin{aligned}
& \sum_{j=k}^l G_{H_0}^{E,\sigma}(n, j) V(j) b(j) \\
&= -b(n) - iW \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (l) + iW \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (k-1).
\end{aligned}$$

By Equation (16), we then get

$$\begin{aligned}
(\Phi_\sigma^E b)(n) &= b(n) + \sum_{j \in \mathbb{Z}} G_{H_0}^{E,\sigma}(n, j) V(j) b(j) = b(n) + \lim_{\substack{l \rightarrow \infty \\ k \rightarrow -\infty}} \sum_{j=k}^l G_{H_0}^{E,\sigma}(n, j) V(j) b(j) \\
&= -i \lim_{l \rightarrow \infty} W \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (l) + i \lim_{k \rightarrow -\infty} W \left(G_{H_0}^{E,-\sigma}(\cdot, n), b \right) (k-1)
\end{aligned}$$

This immediately implies Equation (14), using that, for $\pm(j-n) > 0$, one has

$$\begin{aligned} W\left(G_{H_0}^{E,-\sigma}(\cdot, n), b\right)(j) &= W\left(\mp m_{H_0, \pm}^{E, \sigma} i\left(W_{H_0, \pm}^{E, \sigma}\right)^{-1} m_{H_0, \mp}^{E, -\sigma}(n)^*, b\right)(j) \\ &= \pm i m_{H_0, \mp}^{E, -\sigma}(n) \left(\left(W_{H_0, \pm}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H_0, \pm}^{E, \sigma}, b\right)(j) \\ &\xrightarrow{j \rightarrow \pm \infty} \pm i m_{H_0, \mp}^{E, -\sigma}(n) \left(\left(W_{H_0, \pm}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, \pm}^{E, \sigma}, b\right). \end{aligned}$$

Equation (14) then shows that $\Phi_\sigma^E b \in \mathcal{G}(H_0, E)$ since $m_{H_0, \pm}^{E, -\sigma} \in \mathcal{G}(H_0, E)$. Further, by Theorem 3.1.3(h) and ...[TODO] ($|Z| < 1$), we have

$$\left\|G_{H_0}^{E, \sigma}(n, j)\right\| = \left\|(i\omega^{E, \sigma})^{-1} Z_\sigma^{|n-j|}\right\| \leq \|(i\omega^{E, \sigma})^{-1}\| \|Z_\sigma\|^{n-j} \leq \|(i\omega^{E, \sigma})^{-1}\|,$$

which, together with Equation (16), yields

$$\left\|\Phi_\sigma^E b\right\|_\infty \leq \|b\|_\infty + \|(i\omega^{E, \sigma})^{-1}\| \left(\sum_{j \in \mathbb{Z}} \|V(j)\|\right) \|b\|_\infty < \infty,$$

using (13) in the last step. This shows that $\Phi_\sigma^E b \in \mathcal{B}(H_0, E)$.

(b) As in (a), one shows that

$$(\Psi_\sigma^E b)(n) = b(n) - \sum_{j \in \mathbb{Z}} G_H^{E, \sigma}(n, j) V(j) b(j). \quad (17)$$

Since $b \in \mathcal{B}(H_0, E)$, one has

$$Vb = Vb + (H_0 - E)b = -((E - V) - H_0)b.$$

One then proves (15) exactly as done in (a) by replacing H_0 with H , H with H_0 and E with $E - V$. Equation (15) then shows that $\Psi_\sigma^E b \in \mathcal{G}(H, E)$ since $m_{H, \pm}^{E, -\sigma} \in \mathcal{G}(H, E)$.

(c) By definition, one has

$$W\left(m_{H_0, \pm}^{E, \sigma}, m_{H_0, \mp}^{E, -\sigma}\right) = \left(W_{H, \pm}^{E, \sigma}\right)^*$$

and, by ...[TODO], we have

$$W\left(m_{H_0, \pm}^{E, \sigma}, m_{H_0, \pm}^{E, -\sigma}\right) = 0.$$

By Equation (14) and Equation (15), we then get

$$\begin{aligned} (\Psi_\sigma^E \circ \Phi_\sigma^E) b &= m_{H, -}^{E, -\sigma} \cdot \left(\left(W_{H, +}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H_0, +}^{E, \sigma}, m_{H_0, -}^{E, -\sigma}\right) \left(\left(W_{H_0, +}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, +}^{E, \sigma}, b\right) \\ &\quad + m_{H, -}^{E, -\sigma} \cdot \left(\left(W_{H, +}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H_0, +}^{E, \sigma}, m_{H_0, +}^{E, -\sigma}\right) \left(\left(W_{H_0, -}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, -}^{E, \sigma}, b\right) \\ &\quad + m_{H, +}^{E, -\sigma} \cdot \left(\left(W_{H, -}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H_0, -}^{E, \sigma}, m_{H_0, -}^{E, -\sigma}\right) \left(\left(W_{H_0, +}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, +}^{E, \sigma}, b\right) \\ &\quad + m_{H, +}^{E, -\sigma} \cdot \left(\left(W_{H, -}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H_0, -}^{E, \sigma}, m_{H_0, +}^{E, -\sigma}\right) \left(\left(W_{H_0, -}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, -}^{E, \sigma}, b\right) \\ &= m_{H, -}^{E, -\sigma} \cdot \left(\left(W_{H, +}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, +}^{E, \sigma}, b\right) + m_{H, +}^{E, -\sigma} \cdot \left(\left(W_{H, -}^{E, \sigma}\right)^{-1}\right)^* W\left(m_{H, -}^{E, \sigma}, b\right) = b, \end{aligned}$$

using ...[TODO] in the last step.

(d) By (c), the map

$$\Phi_\sigma^E : \mathcal{B}(H, E) \rightarrow \mathcal{B}(H_0, E)$$

is injective. Since $E \in \mathfrak{D}'$, by ...[TODO], we have

$$\dim \mathcal{B}(H, E) = \dim \mathcal{B}(H_0, E), \quad (18)$$

which implies that Φ_σ^E is an isomorphism. Now, let $b \in \dim \mathcal{B}(H_0, E)$. Then, there exists $b' \in \mathcal{B}(H, E)$ such that $b = \Phi_\sigma^E b'$. By (c), we then have

$$\Psi_\sigma^E b = (\Psi_\sigma^E \circ \Phi_\sigma^E) b' = b' \in \mathcal{B}(H, E).$$

This shows that $\Psi_\sigma^E(\mathcal{B}(H_0, E)) \subseteq \mathcal{B}(H, E)$. Further, by (c), the map

$$\Psi_\sigma^E : \mathcal{B}(H_0, E) \rightarrow \mathcal{B}(H, E)$$

is surjective, which, by (18) implies that it is an isomorphism. $\square$

4 The scattering matrix

In this section, given the Jost solutions constructed previously, we will construct the scattering matrix. It will turn out that this is possible precisely for the energies E for which the spaces of bounded generalized eigenfunctions of H_0 and H have the same dimension. We will show that this is the case for all but countably many energies in the spectrum. Finally, we show that, for every energy in which the scattering matrix is defined, it is unitary and continuous/analytic. **Improve this text. Unitary in the spectrum. Real analytic? Do you extend to the complex plane? Define or recall generalized eigenfunctions. Also in the next text.**

4.1 Construction of the scattering matrix

To construct the scattering matrix, we need bounded Jost solutions with the same asymptotic behavior as a given function in $\mathcal{B}(H_0, E)$. By Lemma 2.2.2(b), we have

$$\mathcal{B}(H_0, E) = \text{span}\{\phi_+^{E, \sigma} e_k : \sigma \in \{\pm\}, k \in J^E\}.$$

Together with Corollary 2.2.5(a), this allows us to make the following definition:

Definition 4.1.1. Let $E \in [-2a_1, 2a_1] \cap \mathbb{C}_A \setminus D = [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$ and $\sigma, \tau \in \{\pm 1\}$. For $k \in J^E$, we define $b_{\tau, k}^{E, \sigma}$ to be the unique function in $\mathcal{B}(H, E)$ such that

$$\lim_{n \rightarrow \tau\infty} \phi_+^{E, \sigma}(n) \sqrt{\langle e_k | \omega^{E, -} e_k \rangle}^{-1} e_k - b_{\tau, k}^{E, \sigma}(n) = 0.$$

(See Definition 2.3.5 for the definition of $\omega^{E, -}$.) Further, we define $b_\tau^{E, \sigma} \in \mathcal{G}_{|J^E|}(H, E)$ by

$$b_\tau^{E, \sigma}(n) := \left(b_{\tau, k}^{E, \sigma}(n) \right)_{k \in J^E} \in \mathcal{M}_{L \times |J^E|}(\mathbb{C})$$

and define $\nu^E \in \mathcal{M}_{L \times |J^E|}(\mathbb{C})$ by

$$\nu_{jk}^E = \delta_{jk} \cdot \sqrt{\langle e_k | \omega^{E, -} e_k \rangle}^{-1}.$$

Remark 4.1.2. (a) The factor $\sqrt{\langle e_k | \omega^{E,-} e_k \rangle}^{-1}$ was chosen in a way such that the Wronskian of the functions $b_{\tau,k}^{E,\sigma}$ has a simple form as we will see in Lemma 4.1.5, which will make further calculations easier. Notice that the term $\langle e_k | \omega^{E,-} e_k \rangle$ is strictly positive and we take above the square root to be positive. Next we argue that the radicand is strictly positive: the fact that it is positive follows from Proposition 2.1.4(c), using that

$$\langle e_k | \omega^{E,-} e_k \rangle = -ia_k [z_{-,k} - z_{-,k}^{-1}] = -ia_k [z_{-,k} - \overline{z_{-,k}}] = 2a_k \operatorname{Im} z_{-,k} \geq 0$$

(we recall that $|z_{-,k}| = 1$, because $k \in J^E$, and therefore $z_{-,k}^{-1} = \overline{z_{-,k}}$). Now we prove that $\langle e_k | \omega^{E,-} e_k \rangle$ cannot be zero. We assume the contrary. We recall that $a_k > 0$ because A is positive an invertible. Then we assume that $\operatorname{Im} z_{-,k} = 0$. Using again the fact that $k \in J^E$, i.e. $|z_{-,k}| = 1$, we obtain that $z_{-,k} = \pm 1$ and therefore

$$E = a_k [z_{-,k}(E) + z_{-,k}(E)] = \pm 2a_k.$$

This contradicts that fact that $E \in \mathbb{C}_A$, which implies that $E \notin \pm\sigma(2A)$.

(b) By definition, we have

$$\lim_{n \rightarrow \tau\infty} \phi_+^{E,\sigma}(n) \nu^E - b_\tau^{E,\sigma}(n) = 0.$$

(c) By definition, we have

$$(\nu^E)^* \omega^{E,-} \nu^E = \mathbf{1}_{|J^E|}.$$

Further, by Proposition 2.1.4(f), for $\sigma \in \{\pm\}$, one has

$$z_{\sigma,k} = z_{-\sigma,k}^{-1} \quad \forall k \in J^E,$$

which, implies

$$Z_\sigma = Z_{-\sigma}^{-1}. \tag{19}$$

We therefore get

$$\omega^{E,+} = -iA [Z_+ - Z_+^{-1}] = -iA [Z_-^{-1} - Z_-] \nu^E = -\omega^{E,-},$$

which yields

$$(\nu^E)^* \omega^{E,\sigma} \nu^E = -\sigma \mathbf{1}_{|J^E|}.$$

We start by calculating the Wronskian of the above-defined generalized eigenfunctions. We get the following analog of Lemma 2.3.7:

Lemma 4.1.3. *sign problem* Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$ and $\sigma, \tau \in \{\pm\}$. Then

$$W(b_\tau^{E,-\sigma}, b_\tau^{E,\sigma}) = 0 \quad \text{and} \quad W(b_\tau^{E,\sigma}, b_\tau^{E,\sigma}) = -\sigma \mathbf{1}.$$

Proof. By (19), we have

$$\phi_+^{E,\pm\sigma} \nu^E = \phi_\mp^{E,-\sigma} \nu^E = \phi_\mp^{\overline{E},-\sigma} \nu^E.$$

By Remark 4.1.2(b), this yields

$$\begin{aligned}
W(b_\tau^{E,\pm\sigma}, b_\tau^{E,\sigma}) &= \lim_{n \rightarrow \tau\infty} W(b_\tau^{E,\pm\sigma}, b_\tau^{E,\sigma})(n) \\
&= \lim_{n \rightarrow \tau\infty} W(\phi_+^{E,\pm\sigma} \nu^E, \phi_+^{E,\sigma} \nu^E)(n) \\
&= W(\phi_+^{E,\pm\sigma} \nu^E, \phi_+^{E,\sigma} \nu^E) \\
&= W(\phi_\mp^{E,-\sigma} \nu^E, \phi_+^{E,\sigma} \nu^E) \\
&= (\nu^E)^* W(\phi_\mp^{E,-\sigma}, \phi_+^{E,\sigma}) \nu^E.
\end{aligned}$$

With Lemma 2.3.7, we then get

$$W(b_\tau^{E,-\sigma}, b_\tau^{E,\sigma}) = (\nu^E)^* W(\phi_+^{E,-\sigma}, \phi_+^{E,\sigma}) \nu^E = (\nu^E)^* \cdot 0 \cdot \nu^E = 0$$

and

$$W(b_\tau^{E,\sigma}, b_\tau^{E,\sigma}) = (\nu^E)^* W(\phi_-^{E,-\sigma}, \phi_+^{E,\sigma}) \nu^E = (\nu^E)^* \omega^{E,\sigma} \nu^E = -\sigma \mathbf{1},$$

using Remark 4.1.2(c) in the last step. $\square$

Since, for $\tau \in \{\pm\}$, $(\phi_\tau^{E,+}, \phi_\tau^{E,-})$ forms a basis of $\mathcal{B}(H_0, E)$, also $(b_+^{E,+}, b_+^{E,-})$ and $(b_-^{E,+}, b_-^{E,-})$ form a basis of $\mathcal{B}(H, E)$ **I do not understand this argument.** . This allows us to make the following definition:

Definition 4.1.4. Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$. Let

$$N_\pm^E, M_\pm^E \in \text{Mat}(|J^E| \times |J^E|, \mathbb{C})$$

This notation was never introduced. In general, notation like this is never introduced (there are several spaces of matrices that are never defined) by the unique matrices such that

$$b_+^{E,+} = b_-^{E,+} M_+^E + b_-^{E,-} N_+^E \quad \text{and} \quad b_-^{E,-} = b_+^{E,-} M_-^E + b_+^{E,+} N_-^E,$$

i.e.

$$b_\sigma^{E,\sigma} = b_{-\sigma}^{E,\sigma} M_\sigma^E + b_{-\sigma}^{E,-\sigma} N_\sigma^E, \quad \sigma \in \{\pm 1\} \quad (20)$$

The so-defined matrices link closely to the Wronskian as the following lemma shows:

Lemma 4.1.5. Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$ and $\sigma \in \{\pm 1\}$. Then the following hold:

- (a) $W(b_{-\sigma}^{E,\sigma}, b_\sigma^{E,\sigma}) = -\sigma M_\sigma^E.$
- (b) $W(b_{-\sigma}^{E,-\sigma}, b_\sigma^{E,\sigma}) = \cancel{\sigma} \sigma N_\sigma^E.$
- (c) $N_-^E = -(N_+^E)^*.$
- (d) $(M_\sigma^E)^* M_\sigma^E = \mathbf{1} + (N_\sigma^E)^* N_\sigma^E$ and M_σ^E is invertible.

Proof. (a) By Lemma 4.1.3, we have

$$\begin{aligned}
W(b_{-\sigma}^{E,\sigma}, b_{\sigma}^{E,\sigma}) &= W(b_{-\sigma}^{E,\sigma}, b_{-\sigma}^{E,\sigma} M_{\sigma}^E + b_{-\sigma}^{E,-\sigma} N_{\sigma}^E) \\
&= W(b_{-\sigma}^{E,\sigma}, b_{-\sigma}^{E,\sigma}) M_{\sigma}^E + W(b_{-\sigma}^{E,\sigma}, b_{-\sigma}^{E,-\sigma}) N_{\sigma}^E \\
&= -\sigma \mathbf{1} \cdot M_{\sigma}^E + 0 \cdot N_{\sigma}^E \\
&= -\sigma M_{\sigma}^E.
\end{aligned}$$

(b) By Lemma 4.1.3, we have

$$\begin{aligned}
W(b_{-\sigma}^{E,-\sigma}, b_{\sigma}^{E,\sigma}) &= W(b_{-\sigma}^{E,-\sigma}, b_{-\sigma}^{E,\sigma} M_{\sigma}^E + b_{-\sigma}^{E,-\sigma} N_{\sigma}^E) \\
&= W(b_{-\sigma}^{E,-\sigma}, b_{-\sigma}^{E,\sigma}) M_{\sigma}^E + W(b_{-\sigma}^{E,-\sigma}, b_{-\sigma}^{E,-\sigma}) N_{\sigma}^E \\
&= 0 \cdot M_{\sigma}^E + \sigma \mathbf{1} \cdot N_{\sigma}^E \\
&= \sigma N_{\sigma}^E.
\end{aligned}$$

(c) By (b), we have

$$N_{-}^E = -W(b_{+}^{E,+}, b_{-}^{E,-}) = -W(b_{-}^{E,-}, b_{+}^{E,+})^* = -(N_{+}^E)^*.$$

(d) By Lemma 4.1.3, we have

$$\begin{aligned}
-\sigma \mathbf{1} &= W(b_{\sigma}^{E,\sigma}, b_{\sigma}^{E,\sigma}) \\
&= W(b_{-\sigma}^{E,\sigma} M_{\sigma}^E + b_{-\sigma}^{E,-\sigma} N_{\sigma}^E, b_{-\sigma}^{E,\sigma} M_{\sigma}^E + b_{-\sigma}^{E,-\sigma} N_{\sigma}^E) \\
&= (M_{\sigma}^E)^* W(b_{-\sigma}^{E,\sigma}, b_{-\sigma}^{E,\sigma}) M_{\sigma}^E + (M_{\sigma}^E)^* W(b_{-\sigma}^{E,\sigma}, b_{-\sigma}^{E,-\sigma}) N_{\sigma}^E \\
&\quad + (N_{\sigma}^E)^* W(b_{-\sigma}^{E,-\sigma}, b_{-\sigma}^{E,\sigma}) M_{\sigma}^E + (N_{\sigma}^E)^* W(b_{-\sigma}^{E,-\sigma}, b_{-\sigma}^{E,-\sigma}) N_{\sigma}^E \\
&= - (M_{\sigma}^E)^* \cdot \sigma \mathbf{1} \cdot M_{\sigma}^E + (M_{\sigma}^E)^* \cdot 0 \cdot N_{\sigma}^E \\
&\quad + (N_{\sigma}^E)^* \cdot 0 \cdot M_{\sigma}^E - (N_{\sigma}^E)^* \cdot (-\sigma \mathbf{1}) \cdot N_{\sigma}^E \\
&= -\sigma (M_{\sigma}^E)^* M_{\sigma}^E + \sigma (N_{\sigma}^E)^* N_{\sigma}^E,
\end{aligned}$$

which implies

$$(M_{\sigma}^E)^* M_{\sigma}^E = \mathbf{1} + (N_{\sigma}^E)^* N_{\sigma}^E \geq \mathbf{1},$$

and therefore M_{σ}^E is invertible. $\square$

Using that the matrices M_{σ}^E are invertible, Equation (20) yields

$$b_{\sigma}^{E,\sigma} (M_{\sigma}^E)^{-1} = b_{-\sigma}^{E,\sigma} + b_{-\sigma}^{E,-\sigma} N_{\sigma}^E (M_{\sigma}^E)^{-1},$$

which is equivalent to

$$b_{-\sigma}^{E,\sigma} = b_{\sigma}^{E,\sigma} (M_{\sigma}^E)^{-1} - b_{-\sigma}^{E,-\sigma} N_{\sigma}^E (M_{\sigma}^E)^{-1}. \quad (21)$$

This motivates the following definition:

Definition 4.1.6. Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$. We define the *scattering matrix*

$$S^E := \begin{pmatrix} (M_+^E)^{-1} & -N_-^E (M_-^E)^{-1} \\ -N_+^E (M_+^E)^{-1} & (M_-^E)^{-1} \end{pmatrix} \in \text{Mat}(2|J^E| \times 2|J^E|, \mathbb{C}).$$

Remark 4.1.7. By (21), we have

$$(b_-^{E,+} \quad b_+^{E,-}) = (b_+^{E,+} \quad b_-^{E,-}) S^E,$$

so S^E is the change-of-basis matrix between the basis $(b_-^{E,+}, b_+^{E,-})$ and $(b_+^{E,+}, b_-^{E,-})$.

4.2 Properties of the scattering matrix

We now prove some properties of the scattering matrix. We start with its unitarity:

Proposition 4.2.1. *Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$. Then S^E is unitary.*

Proof. Setting

$$M := \begin{pmatrix} M_+^E & 0 \\ 0 & M_-^E \end{pmatrix}^{-1},$$

we have

$$S^E = \begin{pmatrix} (M_+^E)^{-1} & -N_-^E (M_-^E)^{-1} \\ -N_+^E (M_+^E)^{-1} & (M_-^E)^{-1} \end{pmatrix} = \begin{pmatrix} \mathbf{1} & -N_-^E \\ -N_+^E & \mathbf{1} \end{pmatrix} M.$$

Using Lemma 4.1.5(c) and (d), this yields

$$\begin{aligned} (S^E)^* S^E &= M^* \begin{pmatrix} \mathbf{1} & -(N_+^E)^* \\ -(N_-^E)^* & \mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{1} & -N_-^E \\ -N_+^E & \mathbf{1} \end{pmatrix} M \\ &= M^* \begin{pmatrix} \mathbf{1} + (N_+^E)^* N_+^E & -(N_+^E)^* - N_-^E \\ -(N_-^E)^* - N_+^E & \mathbf{1} + (N_-^E)^* N_-^E \end{pmatrix} M \\ &= M^* \begin{pmatrix} (M_+^E)^* M_+^E & 0 \\ 0 & (M_-^E)^* M_-^E \end{pmatrix} M = \mathbf{1}_{2|J^E|}. \end{aligned} \quad \square$$

Finally, we show that $E \mapsto S^E$ is continuous outside of D , and we construct an analytic extension. For this, we need the following lemma:

Proposition 4.2.2. *Let $E \in [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D)$, $\tau \in \{\pm\}$ and set*

$$p_\tau^{E,\sigma} := u_\tau^{E,\tau\sigma} \nu^E \in \mathcal{G}_{|J^E|}(H, E) \quad \text{and} \quad q_\tau^E := u_\tau^{E,\sigma} P_{\{1,\dots,L\} \setminus J^E} \in \mathcal{G}_{L-|J^E|}(H, E).$$

Notice that q_τ^E does not depend on σ , this follows from Lemma ?? . If the matrix

$$W(q_{-\tau}^E, q_\tau^E) \in \mathcal{M}_{L-|J^E|}(\mathbb{C})$$

is invertible, then one has

$$b_\tau^{E,\sigma} = p_\tau^E - q_\tau^E W(q_{-\tau}^E, q_\tau^E)^{-1} W(q_{-\tau}^E, p_\tau^{E,\sigma}).$$

Proof. By Lemma ??(a), we have

$$\begin{aligned} b_\tau^{E,\sigma}(n) - p_\tau^E &= b_\tau^{E,\sigma}(n) - u_\tau^{E,\tau\sigma} \nu^E + [\phi_\tau^{E,\tau\sigma}(n) - \phi_+^{E,\sigma}(n)] \nu^E \\ &= b_\tau^{E,\sigma}(n) - \phi_+^{E,\sigma}(n) \nu^E + [\phi_\tau^{E,\tau\sigma}(n) - u_\tau^{E,\tau\sigma}(n)] \nu^E \xrightarrow{n \rightarrow \tau\infty} 0, \end{aligned}$$

using Remark 4.1.2(b) and Corollary ??(b) in the last step. This, by Corollary 2.2.5(b), implies the existence of matrices $B_\pm^{E,\sigma} \in \mathcal{M}_{(L-|J^E|) \times |J^E|}(\mathbb{C})$ such that

$$b_\pm^{E,\sigma} - p_\pm^{E,\sigma} = q_\pm^E B_\pm^{E,\sigma} \quad \text{i.e.} \quad b_\pm^{E,\sigma} = p_\pm^{E,\sigma} + q_\pm^E B_\pm^{E,\sigma}.$$

Further, by Corollary 2.3.4, we have

$$0 = W(q_\mp^E, b_\pm^{E,\sigma}) = W(q_\mp^E, p_\pm^{E,\sigma} + q_\pm^E B_\pm^{E,\sigma}) = W(q_\mp^E, p_\pm^{E,\sigma}) + W(q_\mp^E, q_\pm^E) B_\pm^{E,\sigma},$$

which implies

$$B_\pm^{E,\sigma} = -W(q_\mp^E, q_\pm^E)^{-1} W(q_\mp^E, p_\pm^{E,\sigma}). \quad \square$$

The presentation is confusing, because we define the scattering matrix (together with other objects) on a set (subset of an interval) and later we say that it is analytic in a set where it is not defined. This wording is confusing. It is important to be precise. Anything that is claimed to be analytic has to be defined on an open set of the complex plane.

Corollary 4.2.3. *Consider the function*

$$S : [-2a_1, 2a_1] \setminus (\sigma(2A) \cup \sigma(-2A) \cup D) \rightarrow \bigoplus_{k=0}^L \mathcal{M}_{2k}(\mathbb{C}), \quad E \mapsto S^E.$$

Then the following hold:

- (a) *If $\tau_A = 1$, then S is continuous.*
- (b) *If V obeys the strict growth condition, there exists a discrete subset*

$$\tilde{D} \subseteq \mathbb{C}_A \setminus (-2a_L, 2a_L)$$

with $D \subseteq \tilde{D}$ such that S can be extended to an analytic function on an open set not intersecting $\tilde{D}$.

Proof. We start by noticing that the function Z_- is holomorphic and therefore, by Definition 2.3.5 and Definition 4.1.1, also the function $E \mapsto \nu^E$ is holomorphic (please specify where there are holomorphic). We now prove the statements:

- (a) If $\tau_A = 1$, then $[-2a_1, 2a_1] = [-2a_L, 2a_L]$ and for every $E \in (-2a_L, 2a_L)$, one has $J^E = \{1, \dots, L\}$ please recall the argument. This implies that $b_\tau^{E,\sigma} = u_\tau^{E,\tau\sigma} \nu^E$, see by Proposition 4.2.2. Using Corollary ??(a), we obtain that it is continuous on E . By Lemma 4.1.5(a) and (b), we conclude that M_σ^E and N_σ^E , and therefore S^E , depend continuously on E .

- (b) If V obeys the strict growth condition, by Corollary ??(c) - for $\sigma, \tau \in \{\pm\}$, the functions $E \mapsto u_\tau^{E, \sigma}$ are holomorphic on O_σ . By Proposition 4.2.2, this implies that the functions $E \mapsto p_\tau^{E, \sigma}$ and $E \mapsto q_\tau^E$ can be extended to holomorphic functions defined on an open set. This set depends on O_σ and on the domain of the square root that appears in ν^E , see Definition 4.1.1 and Remark 4.1.2.

Now, let $\tilde{D}$ be the set of elements $E \in \mathbb{C}_A$ such that $W(q_\mp^E, q_\pm^E)$ is not invertible. Since

$$(P_{\{1, \dots, L\} \setminus J^E})^* P_{\{1, \dots, L\} \setminus J^E} = \mathbf{1}_{L-|J^E|} \in \text{GL}_{L-|J^E|}(\mathbb{C}),$$

the set $\tilde{D}$ is discrete in $\mathbb{C}_A$, this is a consequence of Lemma 2.3.9 ~~lem : DZeroSubeq~~ and the definition of $q_\pm^E$ in Proposition 4.2.2. Further, for $E \in (-2a_L, 2a_L)$, one has $J^E = \{1, \dots, L\}$, which implies $\tilde{D} \subseteq \mathbb{C}_A \setminus (-2a_L, 2a_L)$. By Lemma 2.3.11, we have that $D \subseteq \tilde{D}$.

Proposition 4.2.2 and the above arguments imply that $E \mapsto b_\tau^{E, \sigma}$ can be extended to an analytic function defined in an open set not intersecting $\tilde{D}$. By Lemma 4.1.5(a) and (b), this implies that M_σ^E and N_σ^E and therefore also S^E can be extended to holomorphic functions on open sets. What do you do with the inverse of M_σ^E ? $\square$

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